

Ministry of higher education
Culture and Science City at Oct 6
The High Institute of Computer Science & Information Systems



Graduation Project

AI Motion Creator

Submitted by:

#	ID	Name
1	220429	Ali Mohamed Ali Hassan
2	222124	Habiba Yousry Khalil Ibrahim
3	222881	Raghad Abdulalleem Ahmed Mahmoud
4	221206	Saif Aldin Wessam Mohamed Ezzat
5	220428	Youssef Abdallah Mohamed Mostafa
6	220659	Youssef Abdelmageed Ahmed Abdelmageed

Supervised by:

Dr. Marwa El Anany

Assistant

TA. Omar Amir

Project No: F26_302 **Academic Year: 2025 / 2026**

Acknowledgement

With deepest appreciation, we thank Allah for granting us the strength, patience, and ability to see this project through. We are immensely grateful to Culture and Science City and the Dean of the Higher Institute of Computer Science and the Information System for their unwavering cooperation and encouragement, which significantly aided us in completing this project. Their support and partnership have been an invaluable asset, and we are thankful for the opportunities they provided.

We extend our heartfelt gratitude to Dr. Marwa Al Anany and Eng. Omar Amir for their indispensable support and guidance throughout our graduation project. Their unwavering dedication, expertise, and encouragement have been instrumental in our success. We are truly fortunate to have learned from such accomplished and passionate individuals. Their invaluable contributions as supervisors and mentors played a critical role in our achievements, and we are forever grateful for their steadfast support.

Project Abstract

Motion capture systems are widely used in animation, gaming, healthcare, and human computer interaction. However, traditional approaches often rely on specialized hardware, wearable sensors, or physical markers, which increase cost and limit accessibility. This project presents a web-based markerless motion capture system that captures and analyzes human movement using a standard camera without the need for additional sensors.

The proposed system utilizes computer vision and pose estimation techniques to extract body keypoints from video data and convert them into motion information suitable for digital animation and analysis. By deploying the system in a web and cloud-based environment, the solution ensures platform independence, ease of use, and scalability.

The system design, implementation, and testing demonstrate that accurate motion capture can be achieved using low-cost hardware, making the proposed approach a practical and accessible alternative to traditional motion capture systems.

Table of Contents

Chapter 1: Introduction 1
1.1 Overview 1
1.2 Motivation 2
1.3 Objectives 2
1.4 Aim 3
1.5 Scope 3
1.5.1 Deliverables 3
1.5.2 Boundaries and Limitations 4
1.6 General Constraints 4
1.7 Organization of the Project 5
Chapter 2: Background and Previous Work 6
2.1 Background 6
2.1.1 Motion Capture Overview 6
2.1.2 Early Experiments in Motion Capture 6
2.1.4 Rotoscoping: Capturing Movement for Animation 8
2.1.5 Industrial and Clinical Expansion of Motion Capture 9
2.1.6 The Optical Motion Capture Revolution 9
2.1.7 Motion Capture Goes Mainstream 11
2.1.8 The Performance Motion Capture Revolution 12
2.1.9 Markerless and AI-Driven Motion Capture 13
2.2 Previous Work 14
2.2.1 Technological Foundations of Markerless Motion Capture 15
2.2.2 Professional and Research-Grade Systems 16
2.2.3 The Prosumer and Indie Developer Revolution 19
2.2.4 Cloud-Native and Generative AI Tools 20
2.2.5 Developer Ecosystem and Open-Source Solutions 21
Chapter 3: Planning and Analysis 23
3.1 Planning 23
3.1.1 Feasibility Study and Estimated Cost 23
3.1.2 Gantt Chart 25
3.2 Analysis and Limitations of the Existing System 25
3.4 Analysis of the New System 27

3.4.1 User Requirements 27
3.4.2 System Requirements 28
3.4.3 Domain Requirements 29
3.4.4 Functional Requirements 29
3.4.5 Non-Functional Requirements 32
3.5 Advantages of the New System 33
3.6 User Characteristics 34
4.1 Design and Implementation Constraints 36
4.2 Assumptions and Dependencies 37
4.2.1 Assumptions 37
4.2.2 Dependencies 37
4.3 Risks and Risk Management 38
4.4 Design of the Database (ERD) 39
4.4.1 Entity-Relationship Diagram 39
4.4.2 Mapping of the Entity-Relationship Diagram 40
4.5 Class Diagram 40
4.6 Use Case Diagram 41
4.6.1 Primary Use-Case Diagram 41
4.6.2 Use-Case Scenarios 42
4.7 Activity Diagram 43
4.8 Sequence Diagram 44
4.9 State Diagram 45
Chapter 5: Implementation 46
5.1 Software Architecture 46
5.1.1 Technology Stack 47
5.1.2 Back-End API 48
5.1.3 Database 48
5.1.4 Pose-Estimation Model (MediaPipe) 49
5.2 User Interface 50
5.3 Results and Discussion 57
5.3.1 Results 57
5.3.2 Discussion 58
Chapter 6: Testing 59
6.1 Unit Testing 59

6.2 Integration Testing 60
6.2.1 Sample Input and Output 62
Chapter 7: Conclusion and Future Work 65

References 68
Appendices 71
ملخص المشروع باللغة العربية 72

List of Figures

Figure 2.1: The Horse in Motion (Muybridge, 1878). 7
Figure 2.2: Chronophotography technique (Marey). 7
Figure 2.3: Chronocyclegraph (Gilbreth). 7
Figure 2.4: Rotoscoping (Fleischer patent, 1917). 8
Figure 2.5: First motion-capture suit (Harrison, 1959). 10
Figure 2.6: Example of optical marker systems. 10
Figure 2.7: Performance capture: Gollum, The Lord of the Rings. 12
Figure 2.8: Example of facial tracking dots. 12
Figure 2.9: Microsoft's Kinect. 13
Figure 2.10: Avatar: The Way of Water (underwater capture). 14
Figure 3.1: Project Gantt Chart. 25
Figure 4.1: Entity-relationship diagram of the database. 39
Figure 4.2: Class diagram of the AIMotionCreator platform. 41
Figure 4.3: Primary use-case diagram of the platform. 42
Figure 4.4: Activity diagram: generating an animation. 44
Figure 4.5: Sequence diagram for a process-motion request. 45
Figure 4.6: State diagram of a motion-generation job. 45
Figure 5.1: Three-tier system architecture of the platform. 46
Figure 5.2: Studio Dashboard with account stats and recent activity. 51
Figure 5.3: Workspace hub: choosing a toolset. 51
Figure 5.4: Video Motion Capture: selecting the input video. 52
Figure 5.5: Synthesis controls: export type, motion-solver options, and custom rig. 52
Figure 5.6: Cost summary, Generate Motion, and output actions. 53
Figure 5.7: 3D Motion Preview: source video beside the generated character. 53
Figure 5.8: Text-to-Video workspace with the prompt assistant. 54
Figure 5.9: A generated video result shown inline in the chat. 54
Figure 5.10: Projects page listing generated motion files by format and status. 55
Figure 5.11: Assets library: uploaded source videos. 55
Figure 5.12: Assets library: 3D models and rigs available for retargeting. 56
Figure 5.13: Settings: profile, security, appearance, and notifications. 56

Figure 6.1: Integration test input: selecting and uploading the 2D source video. 62

Figure 6.2: Integration test configuration: export type and motion-solver options. 63
Figure 6.3: Integration test output: the generated 3D animation beside the source video. 63
Figure 6.4: Integration test output: the generated file saved in the Projects library. 64

List of Tables

Table 3.1: Estimated cost of developing and operating the platform (indicative). 24
Table 3.2: Characteristics of the platform's intended users. 34
Table 4.1: External dependencies and their impact. 37
Table 4.2: Risk register and mitigation strategies. 38
Table 4.3: Mapping of entities to relational tables. 40
Table 4.4: Use-case scenario: generate 3D animation from video. 42
Table 4.5: Use-case scenario: register and authenticate. 43
Table 5.1: Software, frameworks, and services used in the implementation. 47
Table 5.2: Principal FastAPI endpoints and the logic they perform. 48
Table 5.3: Supabase database schema used by AIMotionCreator. 49
Table 5.4: Measured processing behaviour of the motion pipeline. 58
Table 6.1: Unit test cases and their results. 59
Table 6.2: Integration test cases and their results. 61

Chapter 1: Introduction

1.1 Overview

Three-dimensional (3D) animation and motion capture are central to modern animation, games, film, virtual and augmented reality, sports, and healthcare. Traditionally, capturing realistic human motion requires expensive equipment: marker suits, multi-camera optical studios, or wearable inertial sensors, together with controlled environments and specialized expertise. These requirements place professional motion capture out of reach for students, educators, independent creators, and small studios.

AI Motion Creator is a markerless motion-capture platform that addresses this gap. A user uploads an ordinary 2D video recorded with a standard camera, and the platform uses artificial intelligence and computer vision to detect the performer's body (and, optionally, hand) movement and reconstruct it as a 3D skeletal animation. The result can be previewed in an interactive 3D viewport, retargeted onto a 3D character, and exported in the standard formats used by animation and game-development tools.

The platform is delivered in two modes that share a single cloud account. In the web mode, the user works entirely in the browser, and storage and processing are performed in the cloud. In desktop mode, the application runs on the user's own computer, keeping data in a local database, and performing processing on the local processor, which lowers the running cost for the user. Both modes use the same underlying motion-capture engine and the same account system, so a user's identity is consistent across the two.

By turning a single ordinary video into editable 3D motion, AI Motion Creator removes the cost, hardware, and complexity barriers of traditional motion capture

and makes 3D content creation accessible, fast, and affordable for a much wider audience.

1.2 Motivation

For independent game developers, small studios, educators, and student creators, producing high-quality 3D animation is difficult and expensive. Professional motion capture demands skilled performers, costly marker or sensor hardware, and complex production environments. The cost in time and money strains small teams, stretches development cycles, and leaves little room for the creative work of editing, blending, and polishing animation.

At the same time, advances in computer vision and deep learning have made it possible to estimate human pose directly from ordinary video, without markers or special cameras. This creates an opportunity to replace expensive capture setups with an accessible, software-only alternative. AI Motion Creator is motivated by this opportunity: to give creators back their time and budget by turning any standard video into usable 3D motion, and to make a capability that was once limited to well-funded studios available to anyone with a camera and a computer.

1.3 Objectives

The primary objective is to design, implement, and test a reliable markerless motion-capture platform that converts an ordinary 2D video into editable 3D motion data without specialized hardware. The specific objectives are:

- Build an AI motion-capture pipeline that detects body and optional hand keypoints from video using a pre-trained pose-estimation model and reconstruct them into a consistent 3D skeletal animation.

- Refine the captured motion with biomechanical processing, inverse kinematics, self-collision handling, and ground contact, to produce stable, realistic results.
- Support flexible output: free (root-moving) or in-place motion, full body or upper/lower body, with export to BVH, FBX, and GLB, and retargeting onto built-in or user-supplied character rigs.
- Provide an intuitive interface with an interactive 3D preview, available as a web application, and as a desktop application that shares one cloud account system.
- Persist user accounts, project history, and outputs securely, in the cloud for the web mode and in a local database for the desktop mode.

1.4 Aim

This project aims to make 3D motion capture available to everyone by providing a fast, intuitive, and affordable tool that turns ordinary video into editable 3D animation. By removing the dependence on expensive equipment and controlled studios, the platform empowers students, educators, independent creators, and small studios to create professional-quality 3D motion for education, experimentation, and production.

1.5 Scope

1.5.1 Deliverables

- A markerless motion-capture engine that converts uploaded 2D video into a 3D skeletal animation.
- Body capture with optional hand and finger capture (both hands, left only, or right only).
- Free motion and in-place (locked-root) motion modes, plus full-body or upper/lower-body capture.

- Export to standard formats, BVH, FBX, and GLB, and retargeting onto a built-in default character (Y Bot) or a user-supplied rig.
- An interactive in-browser 3D preview of the generated motion before download.
- A web application (cloud storage and processing) and a desktop application (local database and processing) that share one cloud account system.
- Secure user accounts with saved project history and stored outputs.

1.5.2 Boundaries and Limitations

- The system processes pre-recorded video. Real-time or live capture is not supported.
- It captures a single person per video, not crowds.
- It captures human body and hand motion, not arbitrary objects or props.
- The performer's body and joints must be reasonably visible. Heavy occlusion, very low quality, or cluttered backgrounds reduce accuracy.
- Processing is CPU-based. Output quality and speed depend on the input video and the available processor.

1.6 General Constraints

Because the system infers 3D motion from a single 2D view, it inherits the known limitations of monocular computer vision:

- Depth ambiguity: a single camera cannot measure true depth directly, so depth is inferred by the AI model and can be imperfect, sometimes causing foot sliding or inaccurate ground contact.
- Occlusion: when a limb is hidden or two body parts overlap, keypoints may be missed or mislocated.
- Temporal instability: because the video is analyzed frame by frame, estimates can fluctuate between frames and make the motion appear to jitter unless smoothed.

- Visual confusion: reflections, transparent surfaces, motion blur, and low contrast between the subject and background can mislead the model.

The pipeline mitigates these issues with temporal smoothing, inverse kinematics, self-collision handling, and ground-contact constraints, but they remain inherent constraints of single-camera capture.

1.7 Organization of the Project

This book is organized as follows:

- Chapter 1 introduces the project, its motivation, objectives, aim, scope, and constraints.
- Chapter 2 reviews the background of motion capture and surveys previous and related work.
- Chapter 3 presents project planning and analysis, including the feasibility study, schedule, and system requirements for both the web and desktop modes.
- Chapter 4 presents the system design, covering constraints, assumptions, risks, the overall architecture, and the UML models.
- Chapter 5 describes the implementation, the software architecture, the user interface, and the results.
- Chapter 6 covers system testing, including unit and integration testing.
- Chapter 7 concludes the project and outlines directions for future work.

Chapter 2: Background and Previous Work

2.1 Background

2.1.1 Motion Capture Overview

Motion capture (MoCap) refers to the process of recording the movement of people or objects and translating that motion into a digital representation. It is widely used in animation, film production, video games, biomechanics, sports, and medical rehabilitation. The core objective of motion capture is to accurately capture human motion and represent it in a form that can be analyzed, reproduced, or animated. The evolution of motion capture reflects a long interplay between scientific curiosity, technological innovation, and creative ambition, beginning with early photographic experiments that sought to make motion visible [5], [8].

2.1.2 Early Experiments in Motion Capture

In 1878, British photographer Eadweard Muybridge published The Horse in Motion, a groundbreaking series of photographs commissioned by Leland Stanford to settle the question of how horses run. Using a row of cameras triggered by threads as the horse passed, Muybridge captured successive frames that revealed each stage of motion. His initial setup involved 12 cameras, later expanded to 24, producing one of the earliest visual records of movement and laying the foundation for modern motion capture [1].

In 1882, French physiologist Étienne-Jules Marey coined the term chronophotograph to describe the recording of motion over time. He developed a photographic gun capable of capturing twelve images per second from a single fixed viewpoint. Unlike Muybridge's sequential frames, Marey displayed all pictures onto a single photographic plate, so the entire path of motion appeared layered together. Both Marey and Muybridge dressed subjects in white against a black background to enhance clarity. However, Marey advanced the method by

dressing subjects in black with shiny markers on joints and connecting them with metal bands, ensuring that only essential movement was recorded [1].

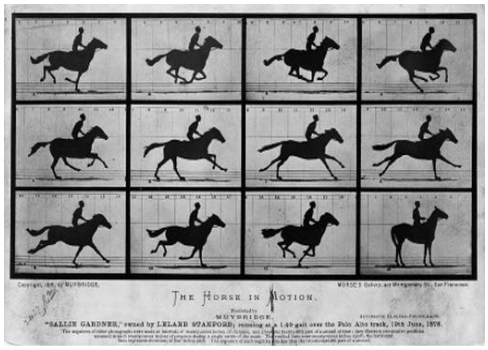


Figure 2.1, The Horse in Motion (Muybridge, 1878).

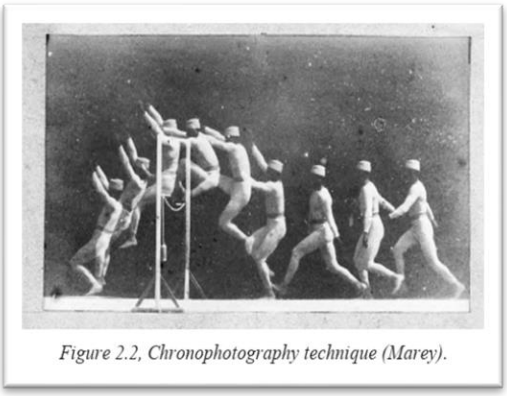


Figure 2.2, Chronophotography technique (Marey).

2.1.3 Motion Analysis in Industrial and Scientific Studies

Between 1911 and 1914, industrial engineers Frank and Lillian Gilbreth advanced motion analysis with the development of chronocyclegraphs, refining the technique during detailed studies for the Herrmann-Aukam company in 1913-1914. Building on Frank Gilbreth's earlier motion-study work, they used long-exposure photographs of workers performing tasks, attaching small incandescent lights to the fingers so movements appeared as continuous trails, while flashing lights created dashed lines along the motion path.



Figure 2.3, Chronocyclegraph (Gilbreth).

This approach not only anticipated modern motion-capture suits with LEDs but also echoed in animation, where Preston Blair later introduced the concept of a line of action to indicate the direction of movement over time. The Gilbreths further refined therbligs, a system of 17 basic motions such as reach, grasp, and release, classified as effective, while others, like position and inspection, were

deemed ineffective. These micromotions represented one of the earliest efforts to break human movement into analyzable components, a principle central to both motion capture and animation [9].

2.1.4 Rotoscoping: Capturing Movement for Animation

The transition from motion analysis to animation was marked by the invention of rotoscoping by Max Fleischer in 1915, patented in 1917. Rotoscoping involved projecting live-action film onto a glass panel, allowing animators to trace each frame and create life-like animated sequences. This technique was first showcased in Fleischer's "Out of the Inkwell" series, featuring Koko the Clown, and later adopted by studios such as Disney for films like Snow White and the Seven Dwarfs (1937). Rotoscoping was one of the earliest forms of motion capture in animation because it allowed animators to transfer real human movement directly into animated characters. Although the process was labor-intensive and required extensive manual work, it introduced the key concept of capturing real motion and applying it to drawn or digital figures. This concept later became the foundation for modern motion-capture technologies [2].

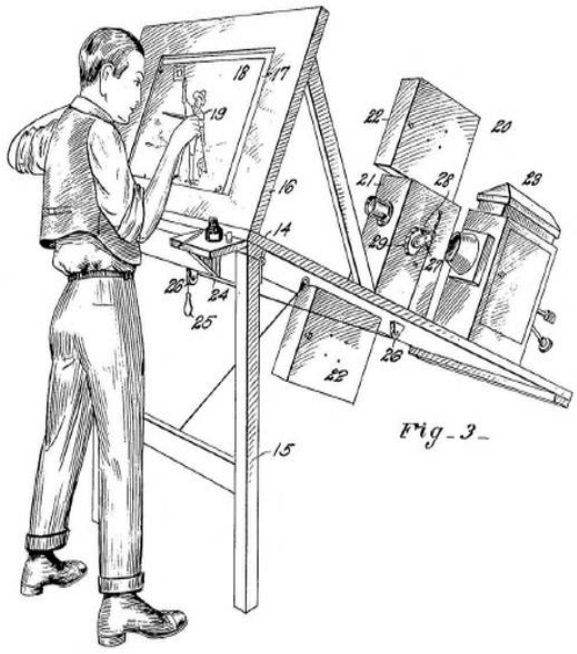


Figure 2.4, Rotoscoping (Fleischer patent, 1917).

2.1.5 Industrial and Clinical Expansion of Motion Capture

The first known motion-capture suit was created by American animator Lee Harrison III in 1959, featuring potentiometers on a bodysuit to record and animate a performer's movements in real time for animation. This early technology, part of his ANIMAC system, converted physical movements into electronic signals and displayed them on cathode-ray-tube (CRT) monitors.

By 1968, Walt Disney Productions, in collaboration with Universal Studios, patented a motion-based animation system derived from Harrison's suit to drive animatronic figures in theme parks, marking one of the earliest commercial applications of motion capture. At the same time, similar systems were applied in medical rehabilitation, military research, and ergonomics, where precise measurement of movement was critical. The 1970s saw the rise of academic research, with universities such as Simon Fraser, MIT, and NYU developing systems for biomechanics and clinical motion analysis. In 1983, Tom Calvert, a professor of Kinesiology and Computer Science, advanced the field with a medical exoskeleton fitted with potentiometers that transmitted patient movement directly to computer animation software. These efforts established motion capture as a powerful tool for analyzing human movement in unprecedented detail [9].

2.1.6 The Optical Motion Capture Revolution

In the 1980s and 1990s, motion capture grew quickly with the arrival of optical marker systems. Instead of heavy suits with wires and mechanical parts, performers now wore tight suits covered with small reflective balls. Several infrared cameras were placed around the stage and worked together to track the balls. At each frame, the cameras calculated the exact 3D position of every marker using triangulation. This gave actors the freedom to move naturally and produced much more detailed data than older exoskeleton systems. Workflows improved as computers became capable of automatically labeling markers, connecting them to digital skeletons, and speeding up calibration [3].



Figure 2.5, First motion-capture suit (Harrison, 1959).



Figure 2.6, Example of optical marker systems.

Early experiments at MIT and the New York Institute of Technology created systems like the Graphical Marionette and Op-Eye, which used infrared lights and multiple cameras to track movement in 3D. By the late 1980s and early 1990s, commercial systems such as VICON, CODA, and Qualisys automated marker tracking and combined it with other data like force plates, muscle activity (EMG), and video. This cut down processing time and made motion capture practical for both clinical and entertainment industries. The main drawback was that optical systems needed a clear view of the markers: if a marker was hidden during fast or complex motion, the data could be lost. Even then, their accuracy and flexibility made them essential tools [3].

At the same time, new projects showed how live performance could drive digital characters. Waldo C. Graphic (1988), created by Jim Henson's Creature Shop with Pacific Data Images, was one of the first real-time animated characters, controlled live by a puppeteer wearing a lightweight sensor exoskeleton. That same year, Silicon Graphics presented Mike Normal at SIGGRAPH, a digital talking head controlled in real time. By 1995, Namco's Soul Edge became one of the first games to use passive optical markers for character animation. Other approaches developed in parallel: magnetic motion capture used sensors to detect changes in magnetic fields but was easily disturbed by interference, while inertial measurement unit (IMU) systems, appearing by the late 1990s, were portable and usable outdoors but suffered from drift that built up over time [8].

2.1.7 Motion Capture Goes Mainstream

By the late 1990s, motion capture shifted from an experimental technique to a mainstream production tool for film and video games. Faster computers, better animation software, and stronger data-processing pipelines made mocap more reliable, scalable, and affordable. It evolved from simply recording movement into a way of translating human performance into digital characters with new levels of realism.

In film, the demand for believable digital characters pushed mocap forward, as filmmakers realized they could preserve an actor's real motion and acting. The most famous example was The Lord of the Rings trilogy, where Andy Serkis's performance as Gollum was captured and transferred directly to the digital character, establishing the idea of performance capture. At the same time, video games embraced mocap to meet rising expectations for realism: sports titles like FIFA and NBA 2K relied heavily on it to replicate authentic athletic motion. Unlike film, games needed motion data that could be blended, looped, and reused dynamically, which led to advanced systems for animation blending and retargeting [7].

Technologically, this period saw major improvements in optical marker systems, higher-resolution cameras, faster frame rates, and more accurate calibration, along with better software for cleanup, retargeting, and pipeline integration. Mocap data could now be mapped onto digital skeletons of different shapes and sizes, allowing one performance to drive multiple characters. By the end of the 1990s, mocap was no longer a novelty but a standard part of digital production, reshaping studio workflows and shifting animators from creating motion entirely by hand to refining and stylizing captured performances [4].

2.1.8 The Performance Motion Capture Revolution

The early 2000s marked a decisive transformation as motion capture expanded into full performance capture, full acting performances of the body, face, and voice. Andy Serkis's portrayal of Gollum in The Fellowship of the Ring (2001) showed that a digital character could carry psychological depth and emotional complexity derived directly from an actor's performance, legitimizing mocap as an artistic tool rather than just a technical shortcut [4].

Pirates of the Caribbean: Dead Man's Chest (2006) featured Bill Nighy as Davy Jones and showcased major advances in facial performance capture. Using sophisticated facial-capture rigs with tracking dots on the actor's face, the production preserved subtle details of Nighy's performance, driving an extremely complex digital face with non-human anatomy. This demonstrated that performance capture could translate authentic human acting onto fantastical characters.



Figure 2.7, Performance capture: Gollum, The Lord of the Rings.



Figure 2.8, Example of facial tracking dots.

Another watershed moment came with Avatar (2009). James Cameron's film integrated full-body mocap, facial capture, and virtual production in one workflow: actors wore head-mounted cameras recording facial expressions and eye movements, while directors used real-time visualization tools to see digital characters inside virtual environments during filming. By The Hobbit (2012), lightweight head-mounted sensor arrays and dense facial marker sets allowed animators to capture micro-expressions such as eye movements, lip tension, and subtle muscle shifts. Rise of the Planet of the Apes (2011) further showed mocap

could work on location, with Andy Serkis's performance as Caesar captured in real outdoor environments.

2.1.9 Markerless and AI-Driven Motion Capture

The first major consumer breakthrough came with Microsoft's Kinect (2010), which used depth-sensing cameras for body tracking in real time without markers or suits. Although its accuracy was limited compared to studio optical systems, it demonstrated that motion capture could be achieved using vision-based algorithms rather than physical markers, opening the field to researchers, hobbyists, and independent developers [10].

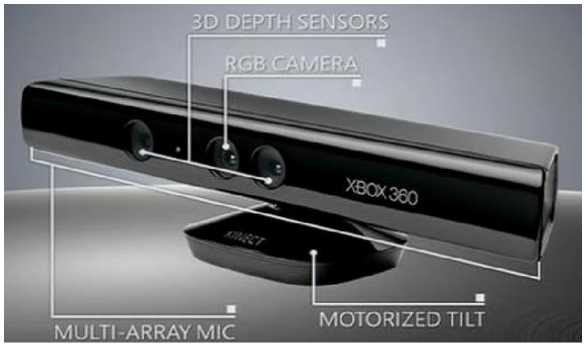


Figure 2.9, Microsoft's Kinect.

As computing power increased, advances in computer vision and deep learning transformed markerless motion capture. Pose-estimation networks trained on large datasets learned to detect human joints and limbs directly from RGB video. By the mid-to-late 2010s, these models could infer 3D human posture from a single camera view, allowing performers to wear everyday clothing and move freely in uncontrolled environments. Simultaneously, IMU-based suits from companies such as Xsens, Rokoko, and Perception Neuron matured into highly portable systems, eliminating the need for multi-camera studios and enabling capture almost anywhere [10].

During the late 2010s, motion capture also converged with real-time rendering engines like Unreal Engine (via Live Link) and Unity, giving rise to virtual production, where mocap data could drive digital characters and environments

live on set. LED volume stages, as popularized by The Mandalorian, allowed actors to perform within immersive digital backdrops, while smartphone-based mocap apps further commercialized access for indie creators. Applications expanded beyond entertainment into VR/AR, robotics, and simulation, and a landmark demonstration of mocap's adaptability came with Avatar: The Way of Water (2022), which pioneered underwater performance capture in a 900,000-gallon tank [6].



Figure 2.10, Avatar: The Way of Water (underwater capture).

Today, artificial intelligence drives motion capture forward, cleaning noisy data, filling gaps, and even generating new motions, while high-fidelity systems capture subtle facial expressions, eye movements, and skin deformations. These advances highlight both promise and difficulty: motion capture has become more accessible, portable, and versatile than ever, yet challenges persist, as markerless systems struggle with occlusion, inertial suits can drift, and high-detail capture remains costly and complex. The ongoing task is to balance accessibility with accuracy, and portability with reliability [8].

2.2 Previous Work

Motion capture is going through its biggest change since digital optical tracking was invented. For over three decades, the industry's gold standard has been defined by marker-based optical systems, highly accurate, requiring high-investment rigs and performers wearing Lycra suits and retro-reflective markers. While this technology remains the benchmark for precision, particularly in high-

end visual effects (VFX) and clinical biomechanics, the rapid maturation of computer vision and artificial intelligence has birthed a formidable challenger: markerless motion capture (MMC) [10].

2.2.1 Technological Foundations of Markerless Motion Capture

To understand the capabilities and limitations of the specific tools detailed later, it helps to first understand the underlying technological architecture that powers modern markerless systems. Unlike marker-based systems that track discrete, high-contrast points in 3D space, markerless systems must interpret the complex, noisy, and variable data of the human form from standard video streams.

The Shift from Depth to RGB-Based Inference

Early markerless technology, like Microsoft Kinect in 2010, relied on RGB-D sensors with infrared depth mapping, groundbreaking for gaming but limited in range, resolution, and outdoor use. The current generation of tools, such as Theia3D and Move.ai, relies on RGB cameras powered by deep neural networks. Instead of measuring depth directly, these systems use massive datasets of annotated human movement to learn how a 2D image of a person correlates to a 3D skeletal pose, allowing the use of standard consumer cameras, from webcams to high-end cinema cameras, by decoupling the capture hardware from the processing software [11].

Top-Down vs. Bottom-Up Pose Estimation

The algorithms driving these tools fall into two categories. Top-down approaches first detect the person in the frame (bounding-box detection) and then run a pose-estimation model within that box. This is highly accurate for single subjects but computationally expensive as crowd size grows, since the pose model must run separately for every detected person. Google's MediaPipe is a prime example. Bottom-up approaches instead detect all potential joints (keypoints) across the whole frame and then use association algorithms (like Part Affinity Fields) to

stitch them into coherent skeletons. OpenPose pioneered this method, which is efficient for crowded scenes because processing time does not scale linearly with the number of people [11], [12].

Volumetric Reconstruction vs. Monocular Prediction

A further distinction is whether a tool is multi-view (volumetric) or single-view (monocular). Volumetric/multi-view systems (e.g., Theia3D, Captury, and Vicon Markerless) require multiple synchronized cameras surrounding the capture volume and triangulate the 3D position of the subject by comparing 2D inferences from different angles, providing true 3D spatial fidelity and robustness against occlusion. Monocular/single-view systems (e.g., DeepMotion, QuickMagic, Rokoko Vision) infer 3D depth from a single 2D video feed. The AI effectively hallucinates the depth from training data. While incredibly accessible, these systems suffer from depth ambiguity, often resulting in foot gliding and inaccurate interaction with the ground plane [13], [27].

2.2.2 Professional and Research-Grade Systems

This sector of markerless technology is the most advanced, built for university labs, clinical gait centers, and AAA game studios. These systems prioritize data integrity, extremely high accuracy, and smooth integration with existing environments, even if they are expensive or harder to set up.

Theia3D (Theia Markerless)

Theia3D has quickly established itself as the leading standard for research-grade markerless motion capture. Unlike many competitors that evolved from gaming, Theia3D was built with a specific focus on biomechanical validity and scientific rigor. As a software-only solution, it works on multi-view video data and effectively takes over the reconstruction stage of a traditional motion-capture pipeline [14], [15], [16], [17].

- Camera configuration: the system requires at least six cameras for reliable tracking (eight or more recommended for complex volumes or multi-person capture), with synchronicity being critical, necessitating genlock-capable machine-vision cameras (e.g., Basler, FLIR) or high-end synchronized setups.
- Computing power: Theia3D relies heavily on deep-learning inference, making the GPU the bottleneck, with performance scaling nearly linearly with GPU capability (for example, an RTX 4090 reaches roughly 657 people/second and an RTX 5090 around 1,202 people/second).
- Ecosystem integration: through the One Ecosystem partnership with Vicon, Theia3D can launch directly from Vicon Nexus, with similar pipelines for Qualisys Track Manager. It operates on an annual subscription for commercial users, with a lifetime license option for academic institutions.

Vicon Markerless (Vanguard System)

While Theia3D partners with Vicon, Vicon has simultaneously developed its own proprietary markerless solution, unveiled fully at GDC and SIGGRAPH 2025, signaling its intent to own the entire hardware-and-software pipeline.

- The Vanguard hardware is a dedicated motion-capture sensor designed to run Vicon's proprietary markerless algorithms at the edge, reducing the latency and bandwidth bottlenecks of transmitting raw video for processing.
- Duplex Mode allows a single set of cameras to capture markerless data (for humans) and marker-based data (for props or rigid bodies) simultaneously, letting an actor perform without a suit while wielding a marker-covered prop, fusing natural movement with sub-millimeter prop precision.
- Real-time capability: via Shōgun Live, markerless data can be streamed into Unreal Engine 5 in real time, targeting the virtual-production market where directors need a live preview of the digital character on the LED wall [18].

The Captury

A German company spun out of the Max Planck Institute, The Captury was one of the first to offer a commercial markerless solution and remains a strong competitor, particularly in live entertainment and VR [19], [20].

- CapturyLive is a turnkey hardware/software solution for real-time tracking, requiring a dedicated vision-processing computer and specific GigE cameras, designed for low-latency applications like broadcast and VR.
- CapturyStudio is a post-processing suite that processes pre-recorded video from any source, offering higher fidelity than Live because non-real-time algorithms can run heavier optimization passes.
- Specifications: for 30 fps video the inherent buffer latency is 33 ms, with total system latency typically 40-80 ms. It is hardware-agnostic (grayscale, GigE, HDMI, SDI) and scales from 4 cameras to 12+ for full 360-degree, multi-actor volumes with simultaneous finger and face tracking.

Simi Shape 3D

Simi Reality Motion Systems is a veteran in sports analytics. Its Simi Shape 3D system is distinct in relying heavily on silhouette extraction combined with biomechanical modeling, and is famously hybrid: researchers can apply markers to specific segments while tracking the rest of the body markerless, fusing the data in Simi Motion software for validation and error correction [21].

- Specialized sports applications include ski jumping (where markers would alter aerodynamics) and baseball/cricket (where markers are banned during live games), using high-speed synchronized video to reconstruct kinematics.
- Validation studies show high reliability for rigid-body tracking and gross body movements, though like all silhouette methods it is sensitive to lighting conditions and background contrast.

2.2.3 The Prosumer and Indie Developer Revolution

While research labs invest in eight-camera arrays, a parallel revolution is occurring in the indie sector. Startups are leveraging consumer hardware, primarily iPhones and GoPros, to build virtual motion-capture studios, sacrificing some spatial precision for massive cost reductions and ease of use.

Move.ai

Move.ai has captured the mindshare of the indie developer community by promising suit-quality data using only mobile phones. Its Move One iOS app allows single-camera capture (limited by monocular depth ambiguity but supporting hand and finger tracking), while its multi-camera system links 2 to 6+ iPhones or GoPros, synchronized over local Wi-Fi peer-to-peer, and uploads the footage to Move.ai's cloud, where proprietary models triangulate the subject, significantly reducing foot sliding and occlusion errors [22], [23].

- Exports: a comprehensive suite of formats including FBX, USD (USDC/USDZ), Blender (.blend), BVH, and C3D, highlighting a target demographic of modern indie pipelines using Blender and Unreal Engine 5.
- Pricing: a Personal plan ($15/month, $126/year) with 2,250 credits/month (~375 seconds of processing, non-commercial), and an Advanced plan ($175/month, $2,100/year) with 37,500 credits/month allowing commercial and multi-camera use.

Rokoko Vision

Rokoko is a hardware-first company (SmartSuits) that has pivoted to include software-only solutions, rebranding "Rokoko Video" to Rokoko Vision. Recognizing that single-camera AI is insufficient for professional depth, its flagship feature is Dual Camera support: pairing any two cameras into a stereo pair drastically improves depth estimation over monocular tools, though it falls

short of the full volumetric coverage of larger setups. The tool is entirely browser-based, lowering the barrier for students and hobbyists [24].

- Pricing: a free Starter tier (limited to 15-second clips), a Plus tier ($20/month, unlimited recording, dual-camera support, and live streaming to engines), and a Pro tier ($50/month, adding face capture and offline capabilities).

2.2.4 Cloud-Native and Generative AI Tools

This category represents the convergence of motion capture and generative AI. These tools are often web-based, accept single video files, and use AI not just to track but to enhance and generate missing data (for example, physics filters and automatic character insertion) [28].

Autodesk Flow Studio

The acquisition of Wonder Dynamics by Autodesk in 2024 shook the industry. Rebranded as Autodesk Flow Studio, this tool is arguably the most advanced "VFX-in-a-box" solution available. When a user uploads a video, the AI tracks the actor (extracting skeletal motion), tracks the camera (solving its movement and focal length), rotoscopes the actor (generating a clean plate and an alpha mask), and composites a CG character into the scene, matching the original lighting and shadows [25].

- Tiers: Lite ($10/month, 1080p, up to 2 people), Standard ($45/month, 4K, 3 people), and Pro ($95/month, high storage, 4 people, unrestricted export formats).
- Exports: the entire scene can be exported as a USD or Blender/Maya/Unreal project, with separate layers for the mocap data (FBX), the clean plate, the alpha mask, and the camera track, crucial for professional compositors.

DeepMotion (Animate 3D)

DeepMotion is a veteran of the browser-based AI mocap space. Its distinguishing technology is a physics engine: after the AI predicts the pose, a physics simulation ensures the character's movements abide by gravity and biomechanical constraints, helping mitigate foot sliding and limb interpenetration. It supports full-body tracking, facial expressions (via webcam), and hand tracking, with foot-locking modes (Auto, Always, Never) [26].

- Pricing: a freemium tier (60 seconds/month, non-commercial), Innovator ($17/month, watermark-free with a commercial license), Professional ($39/month, 4K, 120-second clips), and Studio ($83/month, 8K support).
- Performance: user reviews note that, despite the physics filter, it requires significant cleanup for hero actions and is best used for background characters or quick social-media animations.

2.2.5 Developer Ecosystem and Open-Source Solutions

For those who need to build custom applications or cannot afford commercial licenses, the developer ecosystem offers robust tools.

NVIDIA Maxine AR SDK

For developers with RTX hardware, NVIDIA provides the most powerful local processing SDK, highly optimized for Tensor Cores. On an RTX 4090, its 3D body-pose tracking runs with a latency of roughly 1.37-9.4 ms, orders of magnitude faster than cloud-based solutions, enabling true real-time interaction. It tracks 34 keypoints in 2D and 3D and includes eye-contact (gaze redirection) and high-fidelity face-mesh tracking, powering many VTuber apps and high-end webcam filters [29].

Google MediaPipe

MediaPipe is the workhorse of the mobile and web AR industry. It uses a lightweight, top-down architecture optimized for CPU and mobile-GPU inference, making it the only viable option for running markerless mocap natively on Android or iOS without a server connection. It is also finding traction in clinical research, a 2025 study found its hand-tracking accuracy (RMSE 10.9°) superior to the Leap Motion Controller (14.7°), and, being open source under Apache 2.0, it is free for commercial use, a major advantage over proprietary algorithms [30].

FreeMoCap

FreeMoCap is an ambitious open-source project (led by Jon Matthis) aiming to bring research-grade volumetric capture to the masses for free. It is a hardware-agnostic Python pipeline in which users connect generic webcams via USB hubs. The software uses MediaPipe (or other pose estimators) to find joints in each camera view and then performs a custom triangulation to solve for 3D, bypassing the limitations of single-view MediaPipe. Installation requires a Python environment, and the project is in active development, capable of excellent data but requiring significant technical troubleshooting, making it ideal for researchers on a zero budget who have coding skills [31].

OpenPose (CMU)

Once the king of pose estimation, OpenPose remains a benchmark for accuracy but is hampered by licensing. Its bottom-up approach (Part Affinity Fields) is extremely robust in crowds. However, while it is free for academic use, the commercial license costs $25,000/year, a prohibitive cost that has driven most commercial developers toward MediaPipe or NVIDIA's SDK [32], [33].

Chapter 3: Planning and Analysis

Before any software can be built, it must be planned and analysed. This chapter sets out the planning and analysis carried out for the AIMotionCreator platform, a web-based system that turns an ordinary 2D video into a 3D character animation. It begins with the project plan, including a feasibility study, an estimated cost, and a schedule. Analyses the existing solutions and their limitations. Argues the need for a new system, and finally analyses the proposed system through its user, system, domain, functional, and non-functional requirements.

3.1 Planning

This section establishes whether the project is worth undertaking and how it will be scheduled. It first presents the feasibility study and an estimate of the costs involved, and then sets out the project timeline as a Gantt chart.

3.1.1 Feasibility Study and Estimated Cost

A feasibility study was carried out to confirm that the project could realistically be delivered with the available skills, time, and budget. It considered three dimensions: technical, economic, and operational feasibility.

Technical feasibility. The project is technically feasible because every part of it rests on proven, openly available technology. Reliable body- and hand-tracking can be obtained from established computer-vision libraries, 3D conversion and retargeting can be performed by a mature, scriptable animation engine, and the surrounding web application can be built with widely used web frameworks, a managed database, and standard cloud hosting. The most demanding component, a large language model for the generative features, is made feasible on a modest budget by running it on local hardware and bridging it to the cloud back-end through a secure tunnel, rather than paying for expensive cloud AI hosting. No

unproven research breakthrough is required. The challenge is integration rather than invention.

Economic feasibility. The project is economically feasible because it relies almost entirely on free and open-source software and on the free or low-cost tiers of cloud hosting, database, and storage services. Running the language model locally removes what would otherwise be the largest recurring cost. The only meaningful variable cost is the external generative-video service, which is charged per use and is controlled through the platform's own credit system.

Operational feasibility. The project is operationally feasible because it solves a real problem for a clearly defined audience, independent animators, game-development students, content creators, and small studios who need character animation but cannot justify motion-capture suits or the time required for manual key-framing. The capability is delivered through an ordinary web browser, with an offline desktop version, so the barrier to adoption is very low.

Estimated cost. Because the platform is built on free and open-source software and the free tiers of managed services, its direct development cost is low. Table 3.1 summarises the estimated monthly and one-off costs of building and running the platform at project scale. Figures are indicative and reflect the free/low-cost tiers actually used.

Table 3.1, Estimated cost of developing and operating the platform (indicative).

Item	**Type**	**Estimated Cost**
Front-end hosting (Vercel)	Recurring	Free tier (USD 0 / month)
Back-end hosting (Railway container)	Recurring	Free / low-cost tier (~USD 0-5 / month)
Database, auth & storage (Supabase)	Recurring	Free tier (USD 0 / month)
Domain name (aimotioncreator.tech)	One-off / yearly	~USD 10-15 / year
Generative-video API (Magic Hour)	Variable / pay-per-use	Per credit consumed (metered)
Local AI hardware (LLM host) + tunnel	Existing / one-off	Existing hardware. Ngrok free tier

Open-source software & libraries		Free (USD 0)
Developer effort	One-off	~16 weeks of student development time

3.1.2 Gantt Chart

The project was scheduled over an academic timeline of approximately sixteen weeks, organised into overlapping phases. Planning and requirement gathering came first, alongside research and the feasibility study, followed by system design. Development of the front-end and of the back-end and AI pipeline ran in overlapping parallel, after which the components were integrated and the system was tested and evaluated. Documentation and thesis writing ran continuously throughout. Figure 3.1 presents this schedule as a Gantt chart.

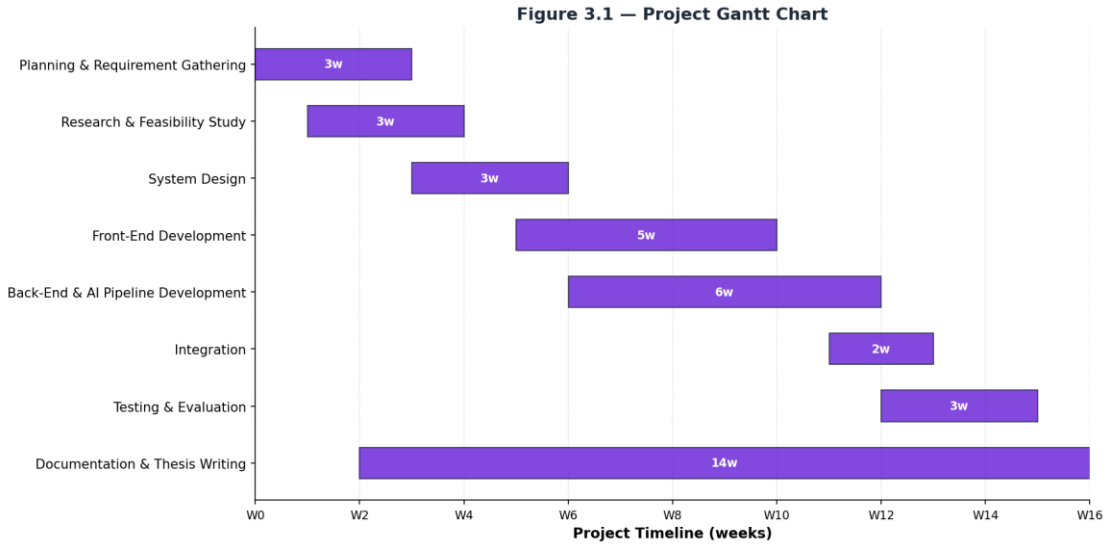


Figure 3.1, Project Gantt chart showing the main phases and their durations.

3.2 Analysis and Limitations of the Existing System

Producing 3D character animation today is possible through several existing approaches, but each has significant limitations for the audience this project serves. Analysing them clarifies the shortcomings a new system must overcome.

- Professional hardware motion capture (marker-based optical systems such as Vicon and OptiTrack, and inertial suits such as Rokoko) delivers very high

accuracy but requires expensive equipment, a dedicated and controlled capture space, and trained operators, putting it far beyond the reach of individuals and small teams.

- Commercial AI cloud services (such as DeepMotion and Move.ai) remove the hardware barrier and reconstruct motion from ordinary video, but they are paid subscription products that run as closed, hosted services, giving the user little control over the process or their data.
- Animation libraries (such as Adobe Mixamo) provide free auto-rigging and ready-made animations, but they supply preset motions only and cannot capture a user's own movement from their own video.
- Open-source components (such as MediaPipe, Blender, and FreeMoCap) are free and powerful but are aimed at technical users: assembling them into a working pipeline demands programming, command-line use, and 3D expertise, and they do not form an integrated application with accounts, storage, preview, and export.
- Manual 3D animation (key-framing by hand in a tool such as Blender) offers complete artistic control but is extremely time-consuming and requires years of specialist skill.

The common thread across these limitations is a trade-off between accessibility and capability: the accurate options are inaccessible, and the accessible options are either closed and paid or demand significant technical skill. None offers a free, easy-to-use, integrated path from a simple video to a finished 3D animation.

3.3 Need for a New System

The analysis above establishes a clear and unmet need. There is a large group of users, independent developers, animation and game-development students, content creators, and small studios, who need 3D character animation but are excluded from the existing options by cost, hardware, closed pricing, or required expertise. What is missing is a single system that brings the accessibility of AI-

based, markerless capture together with the completeness of an integrated application.

A new system is therefore needed that: accepts an ordinary 2D video captured without special hardware. Uses AI to extract the motion automatically. Converts and retargets that motion onto standard 3D formats and custom character rigs. Lets the user preview and export the result, and wraps all of this in a secure, guided, web-based application with user accounts and storage, at little or no cost to the user. AIMotionCreator is designed to meet exactly this need, and the remainder of this chapter analyses its requirements.

3.4 Analysis of the New System

This section analyses the requirements of the proposed system. The requirements are presented at increasing levels of detail: the user requirements describe what users need in plain language. The system requirements translate these into technical terms. The domain requirements capture constraints specific to motion capture and 3D animation. The functional requirements specify precisely what the system must do, and the non-functional requirements describe the qualities it must exhibit. Requirements were gathered by analysing existing solutions, holding informal interviews with potential users, observing current workflows, and reviewing early prototypes.

3.4.1 User Requirements

The user requirements express, in plain language, what the people using the platform need to be able to do:

- Users need to create a personal, secure account and sign in to it.
- Users need to upload a normal 2D video of a person moving, without owning any special hardware.
- Users need the platform to turn that video into a 3D animation automatically, without requiring 3D or programming skills.

- Users need to see a preview of the result before committing to it.
- Users need to apply the captured motion to their own character models.
- Users need to download the result in formats they can use in other tools (game engines and 3D software).
- Users need to find, organise, and manage their uploaded videos and generated animations easily.
- Users need to use additional AI creative tools, such as generating video from text or an image.
- Users need to contact the team for help and to understand the cost of paid features before using them.

3.4.2 System Requirements

The system requirements translate the user needs into technical terms, what the system, and the environment it runs in, must provide:

- The system shall be a web application accessible through a standard modern web browser, with a packaged desktop build for offline use.
- The system shall be built on a three-tier architecture, separating the web front-end, the back-end application logic, and the data layer.
- The back-end shall expose a documented programming interface (API) for processing jobs and shall coordinate the AI motion pipeline.
- The system shall integrate a computer-vision component for body and hand tracking and a 3D engine for conversion and retargeting.
- The system shall use a managed relational database providing authentication, storage, and row-level access control.
- The system shall transmit all data over encrypted (HTTPS) connections and store credentials securely.
- Client devices require only an up-to-date browser and an internet connection. The heavy processing is performed by the server-side pipeline.

- The server environment shall provide sufficient compute to run the AI pipeline, with the language model hosted on local hardware reached through a secure tunnel.

3.4.3 Domain Requirements

The domain requirements capture constraints and expectations that arise specifically from the field of motion capture and 3D character animation:

- The system shall reconstruct a human skeleton, including body and hand joints, from a single-camera 2D video.
- The system shall output motion in recognised, interchangeable industry formats, including a skeleton motion format (BVH) and rigged-model formats (FBX and GLB), so that results can be used in standard 3D software and game engines.
- The system shall support retargeting captured motion onto character rigs whose skeletons differ from the captured skeleton.
- The captured motion shall preserve the correct timing and frame rate of the source video so that the animation plays back at the right speed.
- The system shall account for the inherent limitation that depth and fine detail cannot be fully recovered from a single viewpoint, and shall therefore target plausible, usable motion rather than studio-grade sub-millimetre accuracy.
- Generated outputs shall be valid, well-formed files that open correctly in common 3D tools without manual repair.

3.4.4 Functional Requirements

The functional requirements describe precisely what the system must do. They are organised into six modules, each grouping related features, and every requirement is given a unique identifier so it can be traced through design, implementation, and testing.

User and Account Module, user identity and personal settings.

ID	Requirement
FR-1.1	A visitor shall be able to register a new account using an email address and password.
FR-1.2	A registered user shall be able to log in and log out securely.
FR-1.3	The system shall support multi-factor authentication during sign-in.
FR-1.4	A user shall be able to reset a forgotten password by email.
FR-1.5	A user shall be able to view and update their profile information.
FR-1.6	A user shall be able to manage account preferences, including the application theme.

Motion Capture and Generation Module, the core module that turns an input video into a 3D animation.

ID	Requirement
FR-2.1	A user shall be able to upload a 2D video, or select a previously uploaded video, as the source.
FR-2.2	The system shall automatically extract the body and hand motion from the source video.
FR-2.3	A user shall be able to configure generation options before processing a clip.
FR-2.4	The system shall produce skeleton motion that can be exported in a standard motion format.
FR-2.5	A user shall be able to export a fully rigged 3D model (e.g. FBX / GLB).
FR-2.6	A user shall be able to retarget the captured motion onto a custom character rig.
FR-2.7	A user shall be able to preview the generated animation in an interactive 3D viewport.
FR-2.8	A user shall be able to generate video from a text prompt or an image (AI video generation).

Search and Filter Module, finding content quickly as the library grows.

ID	Requirement
FR-3.1	A user shall be able to browse all of their uploaded assets and generated projects.

FR-3.2	A user shall be able to search assets and projects by name.
FR-3.3	A user shall be able to filter results by type, export format, and status.
FR-3.4	A user shall be able to sort results, for example by date created.

Communication Module, messaging between the user and the platform.

ID	Requirement
FR-4.1	A user shall be able to send a message to the project team through a contact form.
FR-4.2	The system shall deliver submitted contact messages to the team by email.
FR-4.3	A user shall be able to interact with the AI generation tools through a chat-style interface.
FR-4.4	The system shall display clear status and feedback messages while a job is processing.

Transaction and Billing Module, subscription plans and usage credits (payment is simulated in the current version).

ID	Requirement
FR-5.1	A user shall be able to view the available subscription plans and their features.
FR-5.2	The system shall allocate and track a credit balance for each user.
FR-5.3	The system shall deduct credits when a user consumes a paid generation feature.
FR-5.4	A user shall be able to view their current plan and remaining credit usage.

Administration and Dashboard Module, oversight and management functions.

ID	Requirement
FR-6.1	A user shall have a dashboard showing their recent activity and usage statistics.
FR-6.2	A user shall be able to create, rename, and delete their own assets and projects.
FR-6.3	A user shall be able to hide or remove generated outputs from their library.
FR-6.4	An administrator shall be able to oversee user accounts and platform content.

3.4.5 Non-Functional Requirements

The non-functional requirements describe the qualities the system must have, not what it does, but how well it does it.

Performance. Everyday interactions, loading pages, browsing assets, and reading data, should feel immediate, responding within roughly one second under normal use, and the system should remain responsive when many users are active at once. Skeleton motion extraction for a short clip should complete within a few seconds. The heavier rigged-model export is computationally intensive and is expected to take longer, but the system should report progress clearly so the user is never left guessing.

Security. User accounts and data must be protected. All communication must use encrypted connections. Passwords must be stored securely and never in plain form. Multi-factor authentication must be available to strengthen sign-in, and access controls must ensure that each user can see and modify only their own assets, projects, and account data. Sensitive credentials and keys must be kept on the server side and never exposed to the browser.

Usability. The platform must be usable by people who are not 3D-animation experts. The interface should be clean, consistent, and guided, breaking the capture-and-export process into clear steps. It should provide helpful feedback and error messages, work across common screen sizes, and follow good accessibility practice. The target is a System Usability Scale rating in the "good" range or better.

Reliability and availability. The platform should be available whenever users need it, with the cloud-hosted web and data services running continuously. It must handle errors gracefully: a failed or timed-out job should produce a clear message rather than crashing the application, and where a feature depends on an external service, its unavailability should degrade only that feature rather than the whole platform.

Maintainability. The system should be easy to maintain and extend, following a modular design with a clear separation between the front-end, the back-end, and the AI pipeline, so that one part can be changed without disturbing the others. A shared core should drive both the web and desktop versions to avoid duplicated work, and configuration should be kept separate from code so the system can be deployed to different environments without modification.

3.5 Advantages of the New System

Compared with the existing approaches analysed in Section 3.2, the proposed system offers a number of clear advantages:

- Accessibility and low cost. The platform replaces expensive motion-capture hardware and subscription services with an ordinary camera and a web browser, making 3D character animation available to users who were previously priced out.
- No specialist hardware required. Motion is reconstructed from a single 2D video using AI, so there is no need for marker suits, depth cameras, or a controlled capture studio.
- Ease of use for non-experts. A guided, step-by-step workspace breaks the capture-and-export process into clear stages, allowing people without 3D or programming skills to produce usable animation.
- Complete, integrated workflow. Unlike loose open-source components, the platform combines accounts, upload, AI capture, conversion, retargeting, 3D preview, organisation, and export in a single application.
- Standard, interoperable outputs. Animations are exported in recognised industry formats (BVH, FBX, GLB) that work directly in common 3D software and game engines, with support for retargeting onto the user's own character rigs.

- Additional AI creative tools. Built-in text-to-video and image-to-video generation extend the platform beyond motion capture into a broader creative toolkit.
- Cross-platform availability. The same product is delivered both as a web application and as an offline desktop build, so users can work online or locally.
- Security and data ownership. User accounts, uploads, and projects are protected by encrypted connections, secure authentication, and per-user access control, giving users a private, self-contained workspace.

3.6 User Characteristics

The platform is intended for a broad, mostly non-expert audience whose common trait is the need for 3D character animation without access to professional motion-capture resources. Understanding these users, their goals, their technical level, and what they expect, shaped the design of the system. The main categories of user are described below.

Table 3.2, Characteristics of the platform's intended users.

User Type	**Technical Level**	**Goals and Characteristics**
Independent game developers	Moderate (use game engines, limited 3D animation skill)	Need custom character animations for their games quickly and cheaply, without hiring a motion-capture studio. Value standard export formats that drop into their engine.
Animation & game-development students	Beginner to moderate	Want to learn and experiment with real 3D motion using only a camera. Need an easy, guided tool and affordable access while building their skills.
Content creators & hobbyists	Low (little or no 3D background)	Want to bring characters to life for videos and online content quickly. Need a simple, self-explanatory interface and fast results more than fine control.

Small studios & freelancers	Moderate to high	Need a low-cost way to produce animation for client work. Value reliability, reusable assets, and the ability to retarget motion onto supplied character rigs.
Administrator	High (technical operator of the platform)	Maintains the platform, oversees user accounts and content, and ensures the service runs reliably and securely.

Across all of these groups, two characteristics are assumed: users are comfortable operating an ordinary web application in a browser, and they are not expected to have specialist motion-capture or 3D-programming expertise.

Chapter 4: Design

This chapter presents the design of the AIMotionCreator platform. It begins by setting out the constraints that shaped the design, the assumptions and external dependencies it rests on, and the principal risks and how they were managed. It then models the system from several complementary viewpoints: the database is described through an entity-relationship diagram and its relational mapping. The static structure is captured in a class diagram, and the dynamic behaviour is modelled through use-case, activity, sequence, and state diagrams. Together these views form the blueprint that guided the implementation described in the next chapter.

4.1 Design and Implementation Constraints

A number of constraints shaped how the system could be designed and implemented:

- Single-camera input. Motion is reconstructed from one ordinary 2D video, so the design must tolerate the depth ambiguity and occlusion limits inherent to a single viewpoint and target plausible, usable motion rather than studio-grade accuracy.
- Budget and hosting limits. The platform had to be built on free and low-cost service tiers, which constrains available compute, memory, and per-request time and rules out expensive cloud AI hosting.
- Request-time limits. The heaviest stage, converting motion onto a rigged 3D model in the headless 3D engine, can exceed the cloud gateway's request timeout, so the design must support running this stage locally and must report long-running progress to the user.
- Technology constraints. The AI pipeline depends on the Python ecosystem (computer-vision and scientific libraries and a scriptable 3D engine), which fixed the back-end language and framework choices.

- Web and cross-platform delivery. The product must run in a standard browser and also ship as an offline desktop build from a shared code base, which constrains the front-end architecture.
- Security constraints. Because the system manages accounts and uploaded media, the design must enforce encrypted transport, secure credential storage, and strict per-user data isolation throughout.

4.2 Assumptions and Dependencies

The design assumes a number of conditions and relies on several external components.

4.2.1 Assumptions

- Users have a reasonably modern device with an up-to-date web browser and an internet connection.
- Input videos show a reasonably visible single subject, adequately lit and framed, so that the tracker can detect the body.
- Users are comfortable operating an ordinary web application and do not require specialist 3D expertise.
- The free and low-cost service tiers used remain available at the scale of a graduation project.

4.2.2 Dependencies

Table 4.1, External dependencies and their impact.

Dependency	Role	Impact if Unavailable
Vercel	Hosts the web front-end.	Web interface becomes unreachable.
Railway	Hosts the back-end API container.	Processing requests cannot be served.
Supabase	Database, authentication, and storage.	Sign-in and data access fail.
MediaPipe	Body and hand tracking.	Motion cannot be extracted from video.

Blender (headless)	3D conversion and retargeting.	Rigged-model export is unavailable.
Local LLM via tunnel	Generative chat / prompt assistance.	AI assistant features degrade.
Magic Hour API	Text/image-to-video generation.	AI video generation is unavailable.
EmailJS	Delivers contact-form messages.	Contact messages are not sent.

4.3 Risks and Risk Management

The principal risks to the project were identified during design, together with the strategies used to mitigate them. Table 4.2 summarises them by likelihood and impact.

Table 4.2, Risk register and mitigation strategies.

Risk	**Likelihood / Impact**	**Mitigation**
Rigged-model export exceeds the cloud request timeout	High / High	Run the heavy export stage locally or on the desktop build. Report progress. Offer the lighter skeleton export on the cloud.
Poor input video reduces tracking accuracy	Medium / Medium	Guide users on good capture. Target plausible motion. Allow re-running with different options.
External service outage (LLM tunnel, video API, email)	Medium / Medium	Isolate each feature so an outage degrades only that feature, not the whole platform.
Free-tier limits exceeded under load	Medium / Medium	Keep workloads modest. Meter paid generation with a credit system. Leave a path to scale up.
Security breach of user data	Low / High	Enforce HTTPS, hashed passwords, MFA, and row-level per-user access control. Keep secrets server-side.
Scope creep within a fixed timeframe	Medium / Medium	Prioritise the core capture workflow. Simulate billing rather than integrate live payments.

4.4 Design of the Database (ERD)

The platform's data is stored in a managed relational (PostgreSQL) database. Its structure is modelled first as an entity-relationship diagram and then mapped to relational tables.

4.4.1 Entity-Relationship Diagram

Figure 4.1 shows the entity-relationship diagram (ERD). The central entity is the user (held in the authentication service's user records and extended by a Profile). Each user owns many Assets (uploaded videos and generated outputs), many ChatSessions (each containing many ChatMessages), and many stored API keys, and may hide individual assets through HiddenAsset records. The crow's-foot notation shows the one-to-many relationships between these entities.

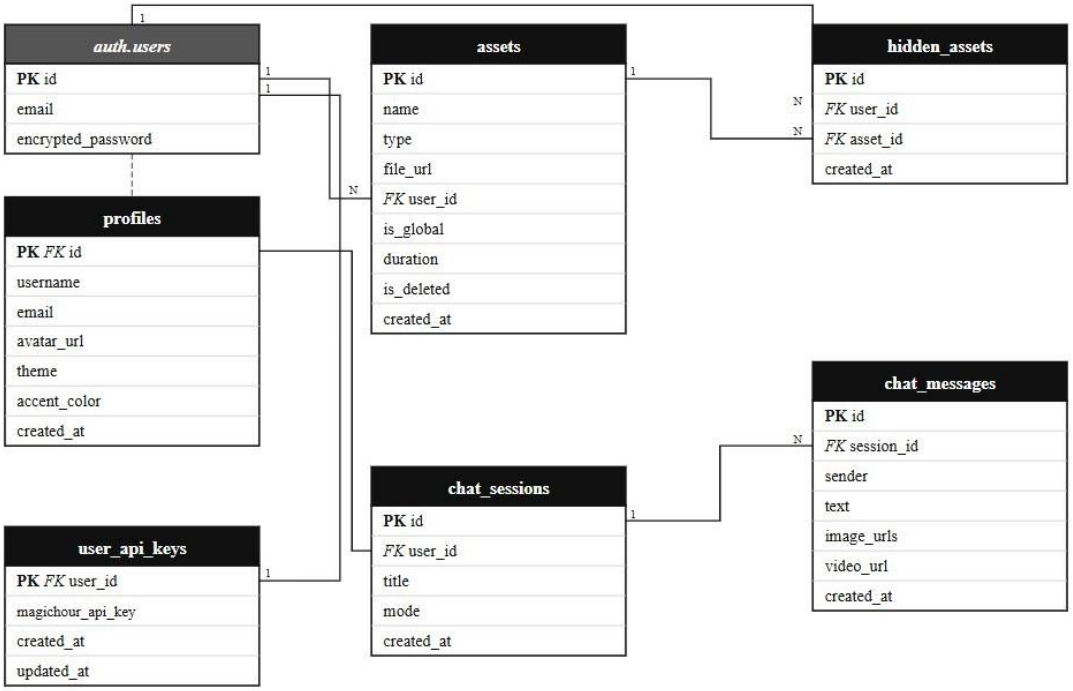


Figure 4.1, Entity-relationship diagram of the database.

4.4.2 Mapping of the Entity-Relationship Diagram

The ERD maps onto six relational tables built on top of the authentication service's user records. Each entity becomes a table, each attribute a column, and each one-to-many relationship is implemented by a foreign key on the "many" side referencing the owning record. Primary keys are UUIDs. Table 4.3 summarises the mapping.

Table 4.3, Mapping of entities to relational tables.

Entity	Table	Primary Key	Key Foreign Keys
User profile	profiles	id	id → auth.users(id)
Asset	assets	id	user_id → profiles(id)
Hidden asset	hidden_assets	id	user_id → profiles(id), asset_id → assets(id)
Chat session	chat_sessions	id	user_id → profiles(id)
Chat message	chat_messages	id	session_id → chat_sessions(id)
API key	user_api_keys	id	user_id → profiles(id)

Every user-owned table carries the owning user's identifier, which lets the database enforce row-level security so that each user can read and modify only their own rows.

4.5 Class Diagram

The class diagram in Figure 4.2 captures the static structure of the system's main data classes and the operations defined on them. Profile is the central class, associated one-to-many with Asset, ChatSession, and UserApiKey. ChatSession is in turn associated one-to-many with ChatMessage, and Asset is associated one-to-many with HiddenAsset. Each class lists its key attributes and the principal operations it supports, such as creating, retargeting, and exporting an asset, or adding a message to a session.

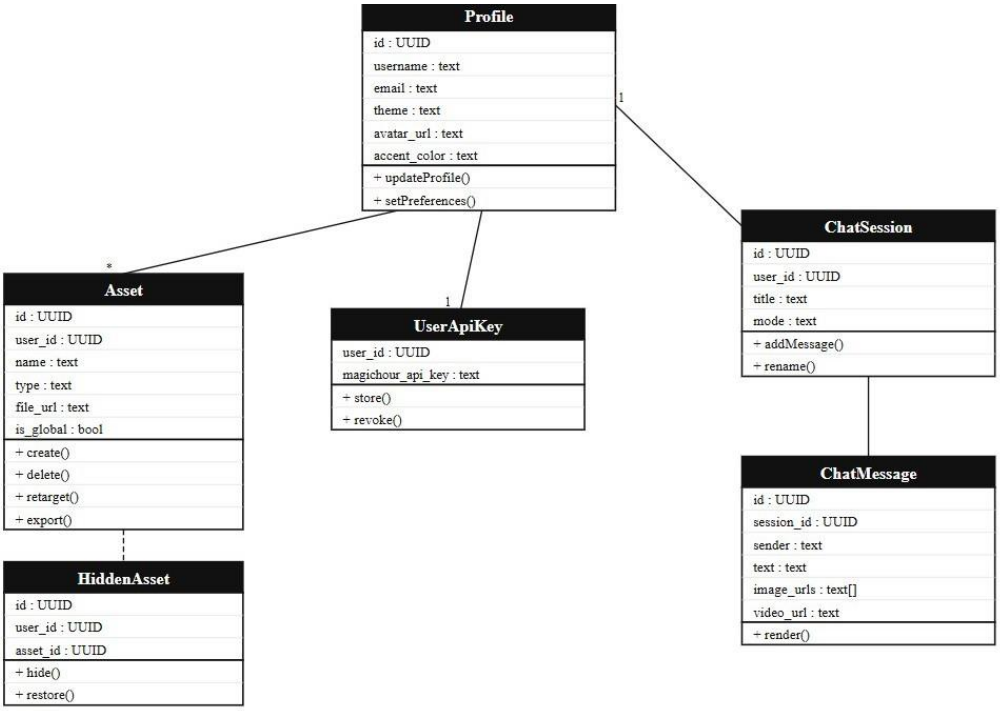


Figure 4.2, Class diagram of the AIMotionCreator platform.

4.6 Use Case Diagram

4.6.1 Primary Use-Case Diagram

Figure 4.3 shows the primary use-case diagram. Three human actors interact with the system: a Visitor, who can register and log in. A Registered User, who has access to the full feature set, uploading video, generating and previewing motion, exporting and retargeting, generating AI video, managing and searching assets and projects, viewing the dashboard, managing billing and credits, and contacting support, and an Administrator, who additionally oversees users and content. Three external actors support the system: the AI Pipeline, the Payment service (currently simulated), and the Email service.

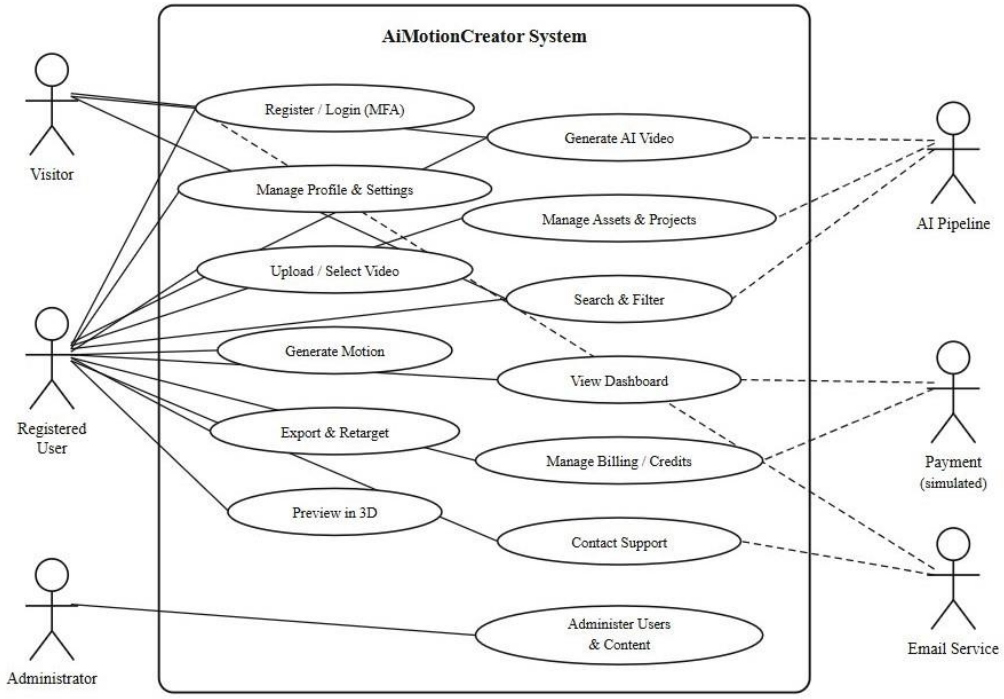


Figure 4.3, Primary use-case diagram of the platform.

4.6.2 Use-Case Scenarios

The two most important use cases are described in scenario form below.

Table 4.4, Use-case scenario: generate 3D animation from video.

Field	**Description**
Use case	Generate 3D animation from video
Actor	Registered User
Pre-condition	The user is logged in and has a 2D video available.
Main flow	1. The user opens the Studio and uploads or selects a video. 2. The user configures the export type and generation options. 3. The system runs the AI pipeline to extract and convert the motion. 4. The system returns the generated animation and a download link. 5. The user previews the animation in the 3D viewport and exports it.
Alternative flow	3a. If processing fails or times out, the system shows a clear error and the user may adjust options and retry.
Post-condition	A generated animation is saved to the user's library and available to download.

Table 4.5, Use-case scenario: register and authenticate.

Field	Description
Use case	Register and authenticate
Actor	Visitor / Registered User
Pre-condition	The visitor has a valid email address.
Main flow	1. The visitor submits an email and password to register. 2. The system creates the account and profile. 3. On later visits the user logs in, optionally completing a second authentication factor. 4. The system issues a signed token and grants access to the user's own data.
Alternative flow	3a. If the user forgets their password, they request a reset link by email.
Post-condition	The user is authenticated and can access only their own assets and projects.

4.7 Activity Diagram

The activity diagram in Figure 4.4 models the flow of actions a user follows to generate an animation. After logging in, the user uploads a video, configures the options, and starts generation. A decision node checks whether processing succeeded: on failure the flow loops back so the user can adjust options and retry. On success the user previews the animation, exports it to a chosen format, and downloads the result.

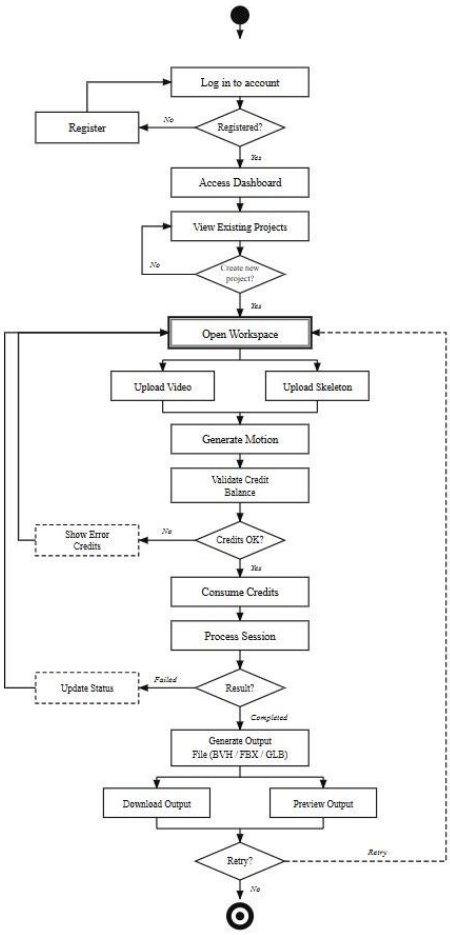


Figure 4.4, Activity diagram: generating an animation.

4.8 Sequence Diagram

The sequence diagram in Figure 4.5 shows the time-ordered exchange of messages between the participants during a motion-generation request. The user uploads a video and options to the web app, which sends a process-motion request to the back-end API. The API invokes the AI pipeline to extract and convert the motion, receives the resulting data and files, and saves an asset record to the database. It then returns the result and a download link to the web app, which displays the 3D preview to the user.

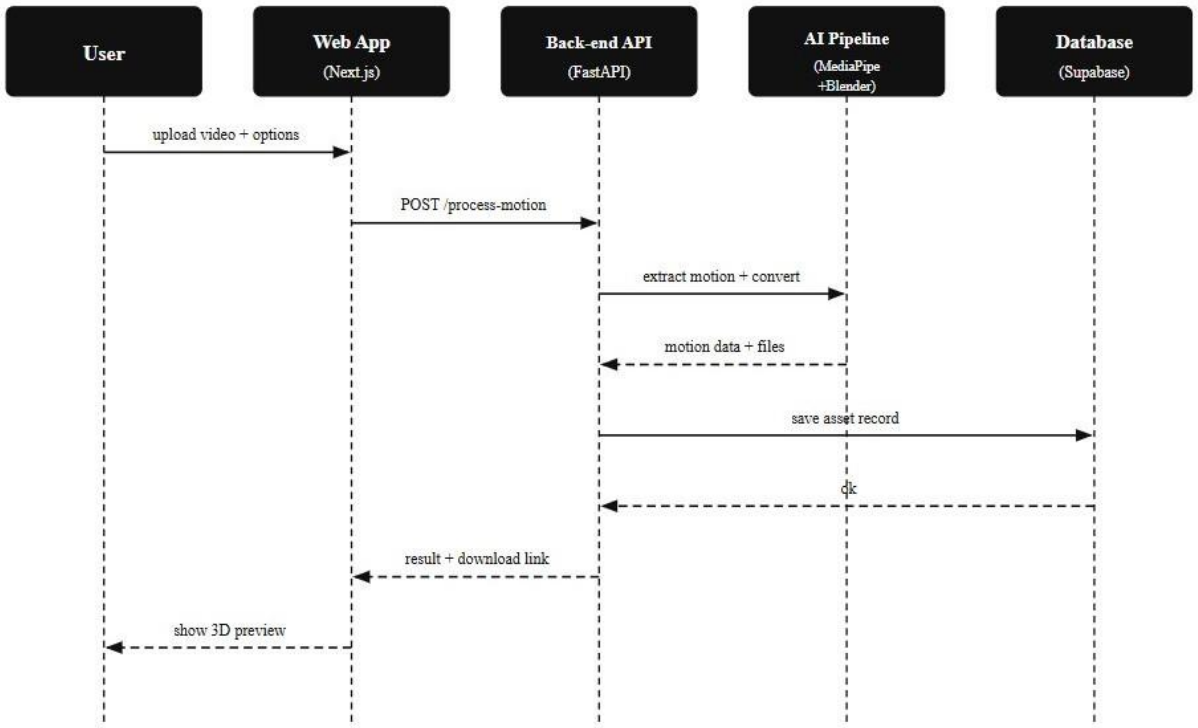


Figure 4.5, Sequence diagram for a process-motion request.

4.9 State Diagram

The state diagram in Figure 4.6 models the lifecycle of a single motion-generation job. A new job begins in the Uploading state. Once the upload completes it moves to Processing. From there it transitions either to Completed on success or to Failed on error. A completed job can be Exported and then reaches its final state, while a failed job can be retried, returning to Processing, or terminated.

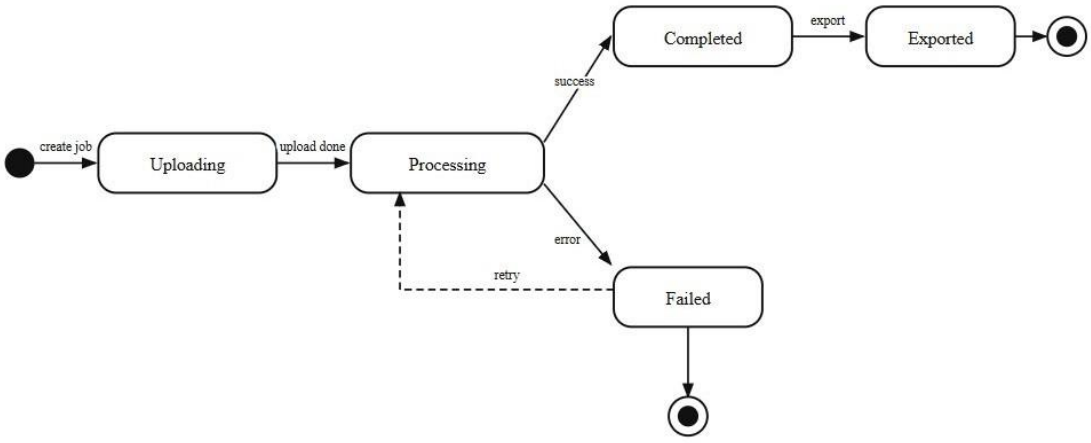


Figure 4.6, State diagram of a motion-generation job.

Chapter 5: Implementation

This chapter describes how the AIMotionCreator platform was implemented and how it performed. It first presents the software architecture, the three-tier structure, the technology stack, the back-end API, and the database. It then walks through the user interface as the user experiences it. Finally, it reports the results obtained from testing the finished platform and discusses what those results reveal about its strengths and limitations.

5.1 Software Architecture

The platform is built on a three-tier architecture that separates the presentation layer (the web and desktop interface), the business-logic layer (the back-end server and AI pipeline), and the data layer (the managed database, authentication, and storage). The same core code base produces both a cloud-hosted web application and an offline desktop application. Figure 5.1 shows the overall architecture and how the layers communicate.

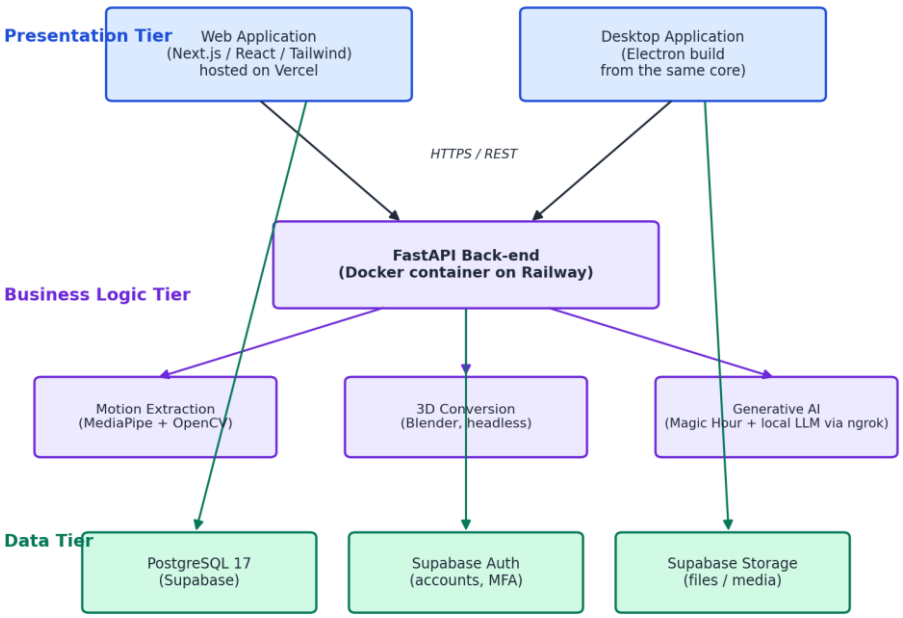


Figure 5.1, Three-tier system architecture of the platform.

5.1.1 Technology Stack

Development was carried out in Visual Studio Code, using TypeScript and React for the client and Python for the server and AI pipeline. Table 5.1 lists the principal languages, frameworks, libraries, and third-party services used across the stack. GitHub is the single source of truth: both the front-end host (Vercel) and the back-end host (Railway) are connected directly to the repository and redeploy automatically whenever the code is updated, giving continuous deployment without a separate build pipeline.

Table 5.1, Software, frameworks, and services used in the implementation.

Category	**Tools / Services**
IDE	Visual Studio Code
Languages	TypeScript, JavaScript, Python 3.10, GLSL (via Three.js shaders)
Front-end framework	Next.js (React, App Router) [38], Tailwind CSS
3D rendering	Three.js [39], interactive WebGL rendering with orbit controls and BVH/FBX/GLB loaders
Back-end framework	FastAPI (Python) [37], Uvicorn ASGI server
AI / motion pipeline	Google MediaPipe (Pose + Hand Landmarker) [34], NumPy [40], SciPy [41], OpenCV [36]
3D conversion engine	Blender (headless) [35] for BVH-to-FBX/GLB export [42] and rig retargeting [43]
Generative AI	Local language model via Ollama (Qwen 2.5). Magic Hour for text/image-to-video
Database / Auth / Storage	Supabase (managed PostgreSQL 17, Supabase Auth, Supabase Storage)
Email delivery	EmailJS for the public Contact form
Source control & CI/CD	Git and GitHub with auto-deploy on push
Hosting	Vercel (front-end, domain aimotioncreator.tech). Railway (back-end, Docker)
Local model tunneling	ngrok (secure tunnel from the cloud back-end to the local model)
Desktop packaging	Electron, esbuild, electron-builder

5.1.2 Back-End API

The back-end is a FastAPI (Python) application that orchestrates the AI pipeline and a router dedicated to AI video generation. Cross-origin requests are enabled so the Vercel-hosted front-end can call the Railway-hosted back-end across domains. Table 5.2 summarises the principal endpoints and the logic each performs.

Table 5.2, Principal FastAPI endpoints and the logic they perform.

Endpoint	Business Logic
POST /process-motion	Saves the uploaded video (and any custom rig), runs the full motion-extraction pipeline, and returns the animation with a download link.
POST /cleanup-motion	Re-imports a generated FBX/GLB into Blender to smooth and validate the animation.
GET /outputs/{file}	Streams a generated file for viewing in the in-browser 3D viewport.
GET /download/{id}	Provides a generated file for download under its original filename.
POST /open-in-blender	Launches Blender on the user's machine with the exported model loaded.
/text-video/*	Wraps the Magic Hour service and the local language model for text/image-to-video generation, including per-user access-key storage.
GET /	Health check confirming that the server and AI pipeline started correctly.

5.1.3 Database

Persistent data is stored in a managed Supabase project built on PostgreSQL 17. The front-end connects directly using the project URL and public key, while the back-end connects through a helper layer that forwards each caller's access token, so Row-Level Security, enabled on every table, restricts every user to their own rows. The live schema consists of the six tables summarised in Table 5.3, while the actual files (videos, rigs, and BVH/FBX/GLB outputs) are written to a

Supabase Storage bucket with only their metadata and public links kept in the database.

Table 5.3, Supabase database schema used by AIMotionCreator.

Table	Purpose / Key Fields
profiles	One row per user (username, email, avatar, theme, accent colour).
assets	Uploaded and generated files (name, type, owner, duration, link, sharing/deletion flags).
hidden_assets	A per-user list of files the user has chosen to hide.
chat_sessions	Text/image-to-video conversations (owner, title, mode).
chat_messages	Individual messages within a conversation (sender, text, generated media).
user_api_keys	Each user's stored access key for the video-generation service.

5.1.4 Pose-Estimation Model (MediaPipe)

The core of the motion-extraction stage is Google's MediaPipe Pose Landmarker, built on the BlazePose GHUM 3D model, together with the MediaPipe Hand Landmarker for finger capture [30], [34]. Choosing a pre-trained, openly licensed model rather than training one from scratch is what makes the project technically and economically feasible.

Model and training data. The pose model is a lightweight convolutional neural network (a MobileNetV2-style architecture) released by Google under the Apache 2.0 licence and offered in three variants, Lite (about 3 MB), Full (about 6 MB), and Heavy (about 26 MB), trading speed for accuracy. According to its model card it was trained and evaluated on real-world smartphone images captured in the wild through a mobile augmented-reality application: roughly 30,000 consented images of people, plus about 85,000 images covering a wide range of fitness poses. Its depth (z) coordinate is not produced from human annotation but is fitted from synthetic data using the GHUM articulated 3D human-shape model, which means the depth is up to scale rather than metric. The

hand model is trained the same way, on consented smartphone images from an augmented-reality application, and returns 21 three-dimensional landmarks per hand together with a handedness score, also grounded in the GHUM hand model.

What it produces. For each video frame the pose model returns 33 body keypoints, covering the head, shoulders, elbows, wrists, hands, hips, knees, ankles, and feet, each with an (x, y, z) position together with visibility and presence scores. The hand model adds 21 keypoints per hand. These keypoints are the raw signal from which the rest of the pipeline reconstructs the skeleton.

How we use it. AI Motion Creator runs the pose landmarker, and, when finger capture is enabled, the hand landmarker, frame by frame over the uploaded 2D video, using the Full or Heavy variant where accuracy matters most. The per-frame keypoints are smoothed over time and converted, using rotation and inverse-kinematics calculations, into a consistent 3D skeleton, which is then exported to the standard BVH, FBX, and GLB formats and retargeted onto the chosen character rig. Because the model tracks a single person and estimates depth only up to scale, the system targets clearly filmed, single-subject video, the same limitations set out in the project's general constraints.

5.2 User Interface

The front-end is a Next.js and React application styled with Tailwind CSS. Public pages share a common navigation bar and footer. Once signed in, the user enters the Studio, a persistent dark-themed layout with a collapsible left sidebar (Dashboard, Workspace, Projects, Assets, Billing, and Settings) and a consistent purple accent. The screens below walk through the Studio interface as the user experiences it.

Dashboard. After signing in, the user lands on a personalised Dashboard (Figure 5.2) that greets them by name and summarises their account through stat cards (capture time, credits remaining, storage used), a credits-usage bar, and a recent-activity table of their latest motion files.

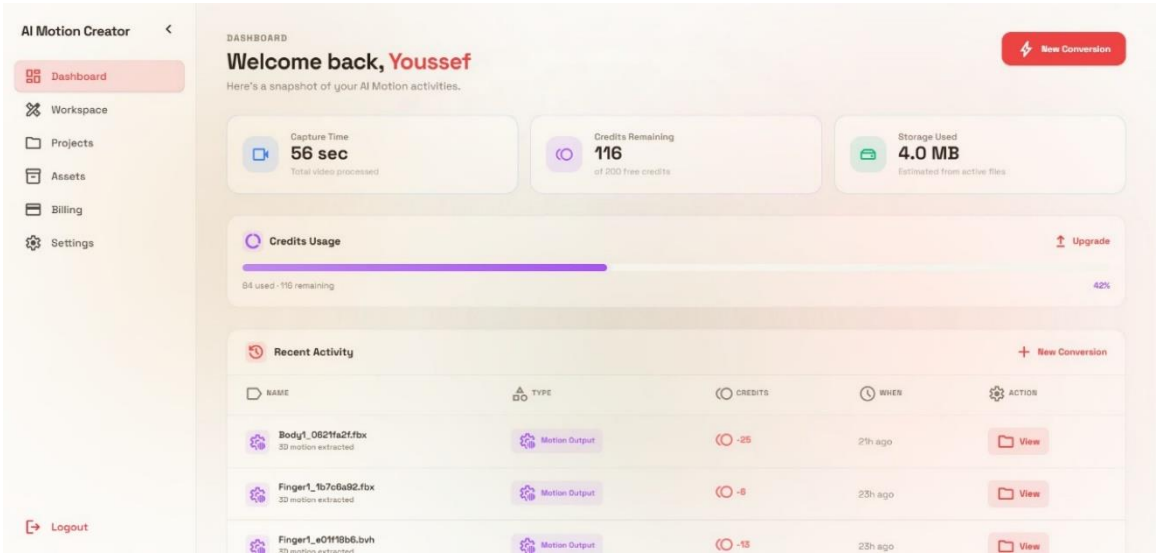


Figure 5.2, Studio Dashboard with account stats and recent activity.

Workspace hub. The Workspace entry screen (Figure 5.3) presents the two production tools as selectable cards, Video Motion Capture and Text-to-Video generation, each opening its own dedicated workspace.

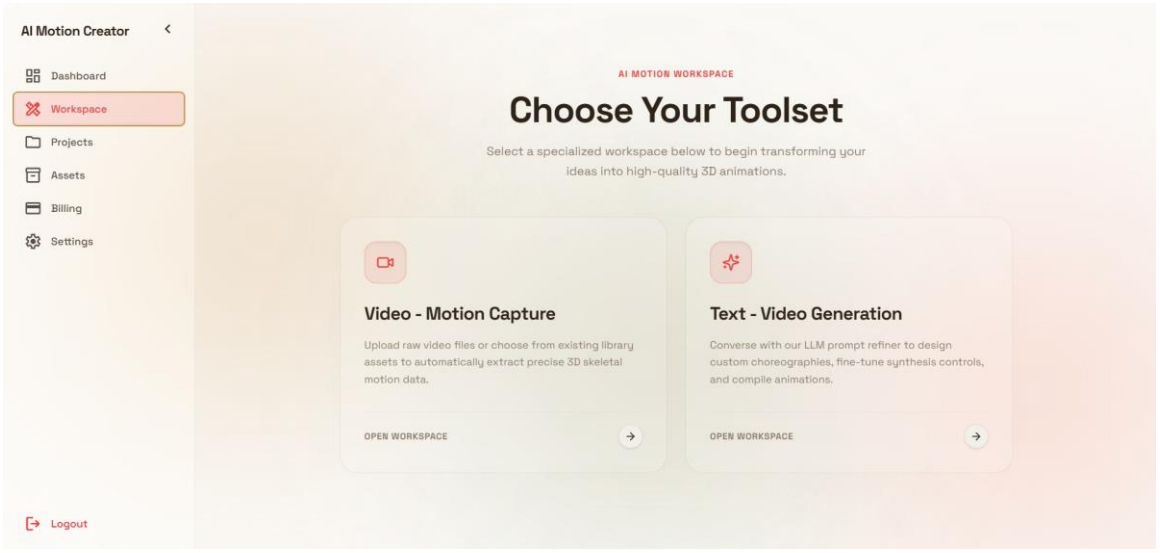


Figure 5.3, Workspace hub: choosing a toolset.

Video Motion Capture, source selection. The motion-capture workspace begins with a Select Video Source step (Figure 5.4), offering a drag-and-drop upload zone or a gallery of previously uploaded and sample videos.

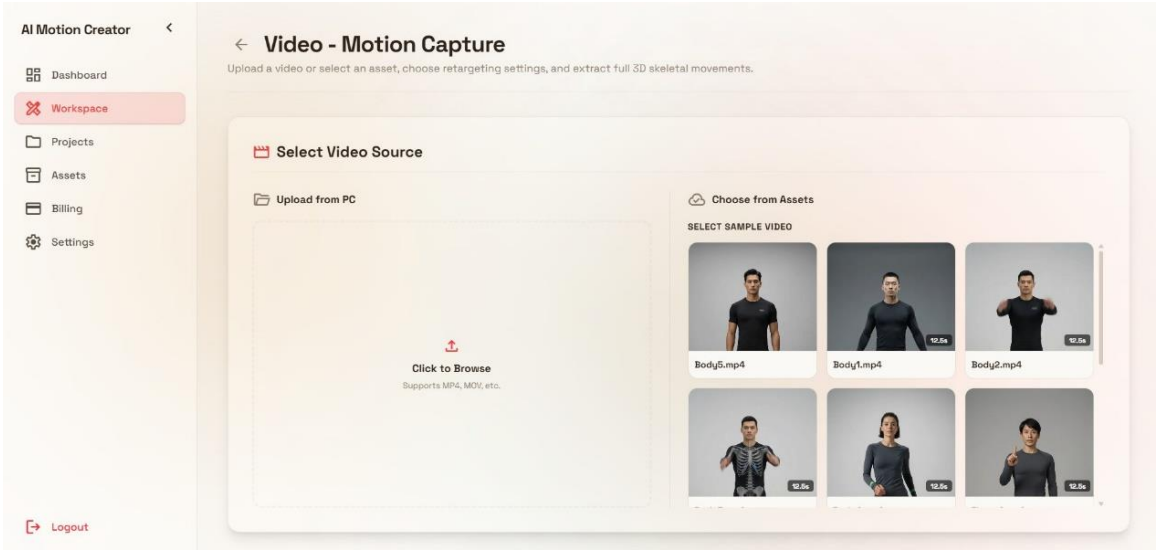


Figure 5.4, Video Motion Capture: selecting the input video.

Video Motion Capture, synthesis controls. The user then sets a clip range and the synthesis controls (Figure 5.5): the export type (BVH, FBX, GLB, or retargeted), the motion-solver options (ground locking, root rotation, upper/lower-body masking, and finger capture), the rest pose, and an optional custom rig to retarget the motion onto.

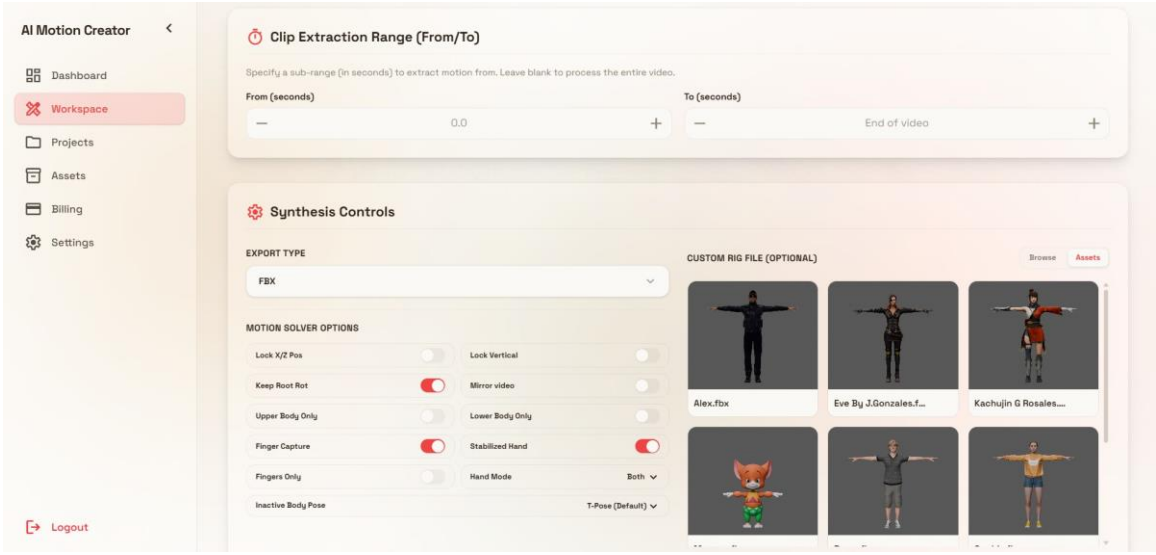


Figure 5.5, Synthesis controls: export type, motion-solver options, and custom rig.

Video Motion Capture, generation. A cost and credit summary (Figure 5.6) reports the video duration, processed length, and credits required. The Generate Motion action then runs the pipeline and enables the preview, render, and download buttons.

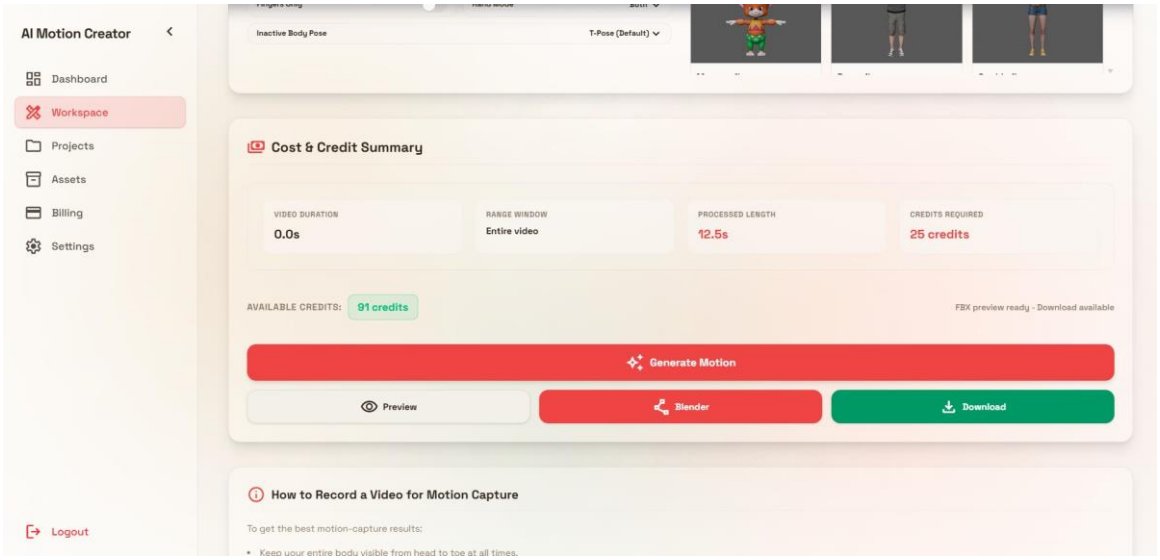


Figure 5.6, Cost summary, Generate Motion, and output actions.

Video Motion Capture, 3D output preview. The result opens in a 3D Motion Preview window (Figure 5.7) that shows the source video beside the generated rigged character in an interactive 3D viewport, with download and open-in-Blender actions.

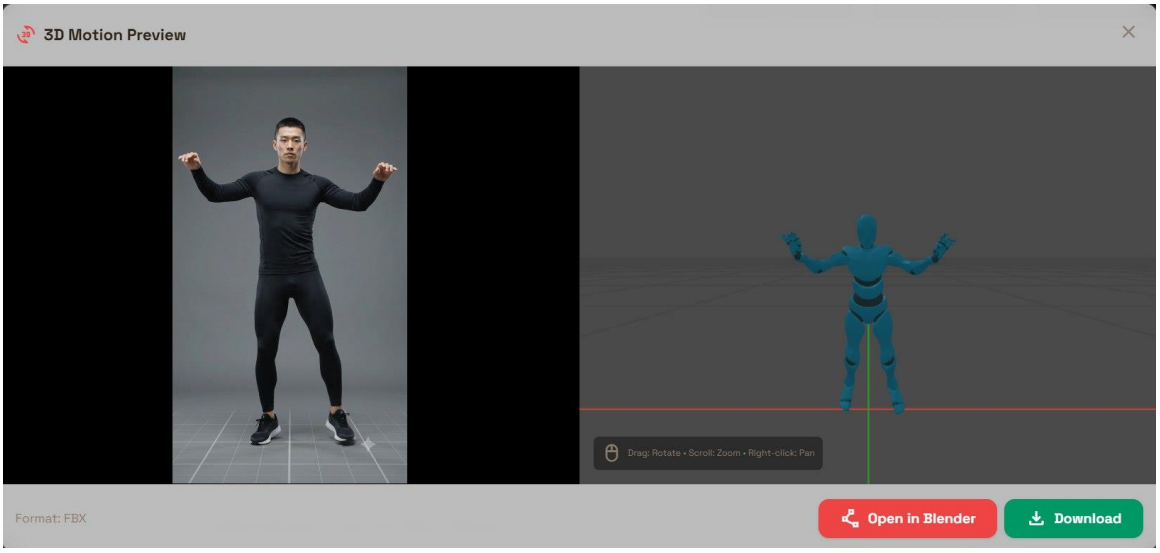


Figure 5.7, 3D Motion Preview: source video beside the generated character.

Text-to-Video, prompt assistant. The second workspace (Figure 5.8) is a chat-assisted generator supporting text-to-image, text-to-video, image-to-video, and storyboard modes. A prompt-assistant panel helps draft the prompt while a video-settings panel configures the model, duration, aspect ratio, resolution, and audio.

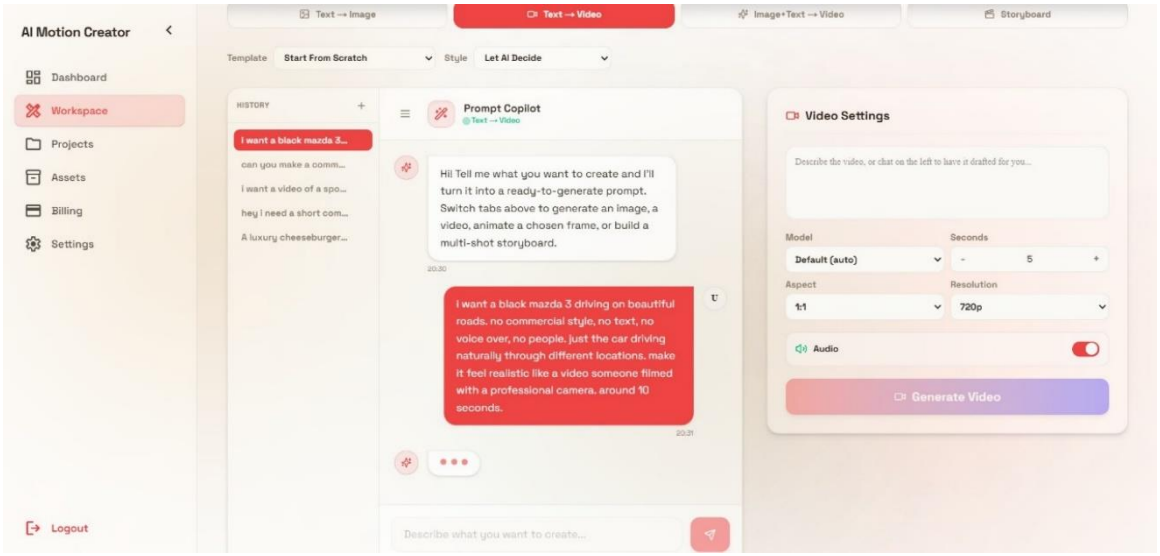


Figure 5.8, Text-to-Video workspace with the prompt assistant.

Text-to-Video, generated result. Once generated through the Magic Hour service, the finished clip plays inline in the chat with its credit cost reported (Figure 5.9), and the user can iterate by continuing the conversation.

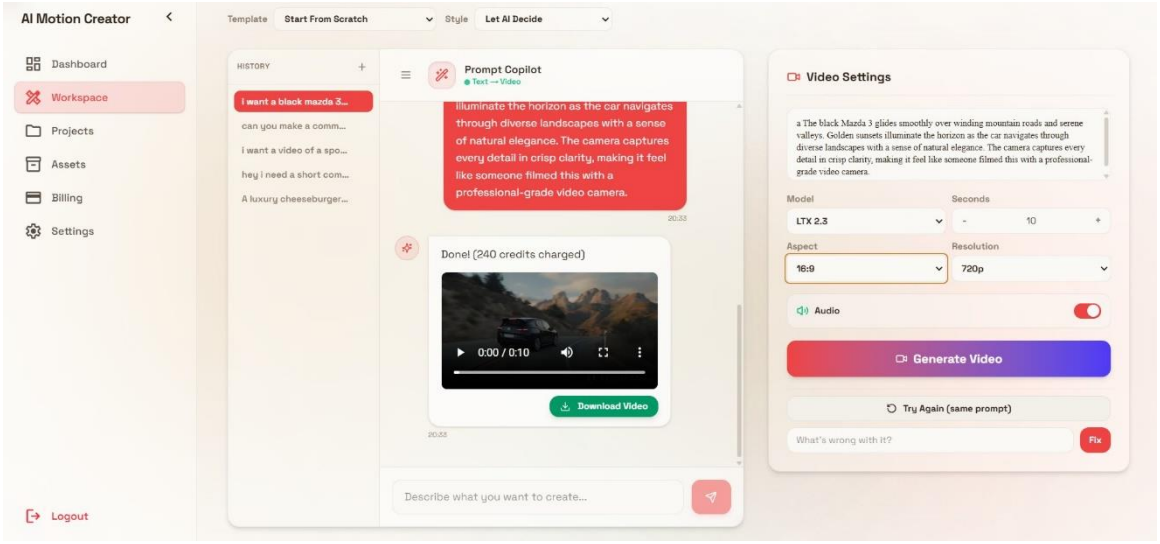


Figure 5.9, A generated video result shown inline in the chat.

Projects. The Projects page (Figure 5.10) lists all generated files in a table filterable by format (All, FBX, GLB/GLTF, or BVH), each row showing a status badge, file size, and date, with preview, download, and share actions.

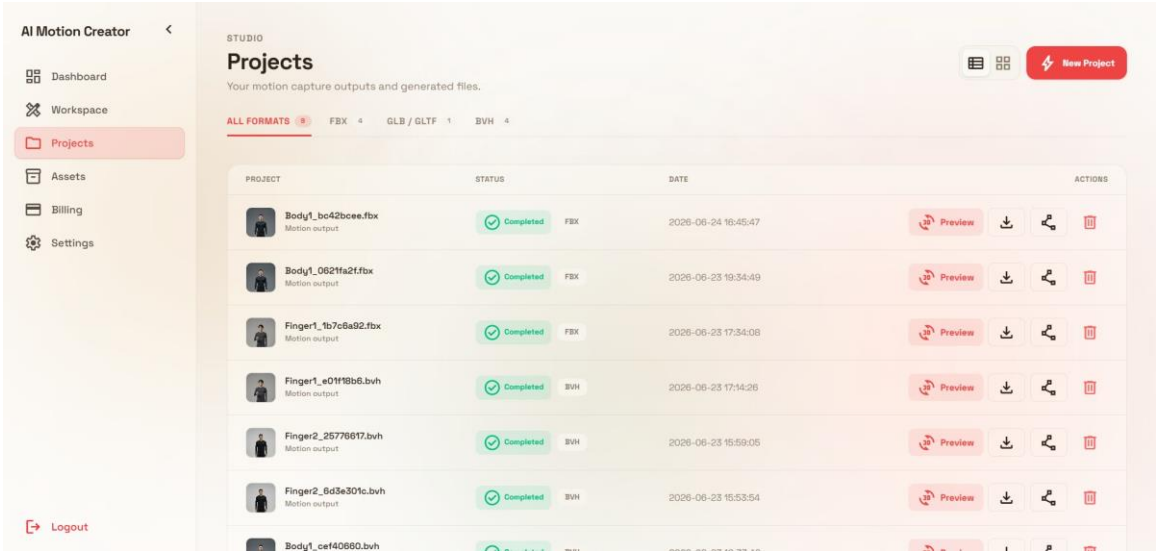


Figure 5.10, Projects page listing generated motion files by format and status.

Assets, videos. The Assets library stores the user's uploads. A drag-and-drop zone accepts new uploads, and a videos grid (Figure 5.11) presents source clips as thumbnails with download and delete controls.

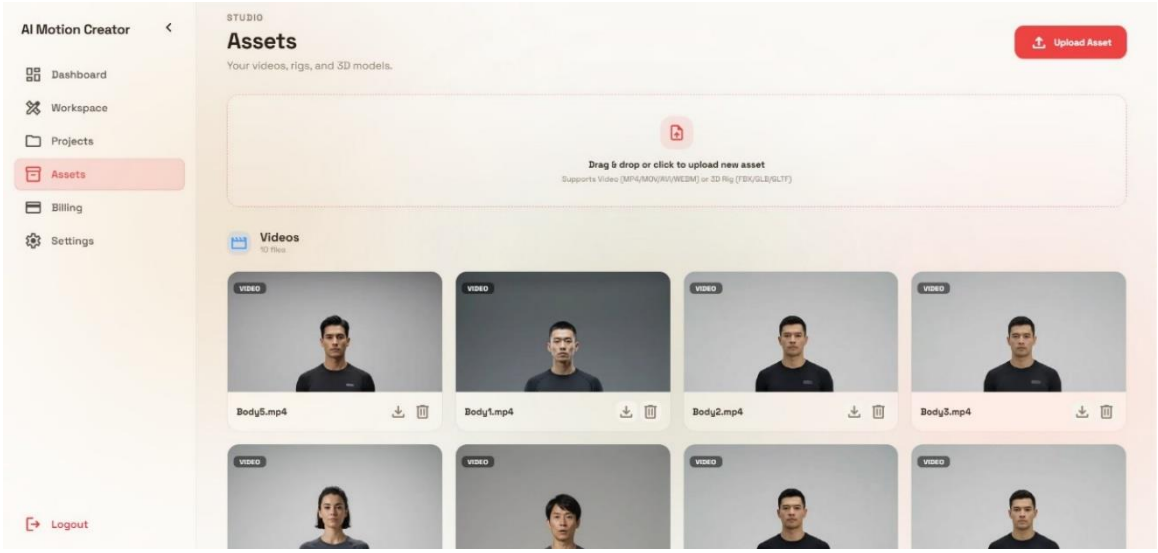


Figure 5.11, Assets library: uploaded source videos.

Assets, 3D models and rigs. A second Assets section (Figure 5.12) lists the 3D character models and rigs (FBX and GLB) that can be selected as retarget targets, each shown as a rendered thumbnail with its file name and format.

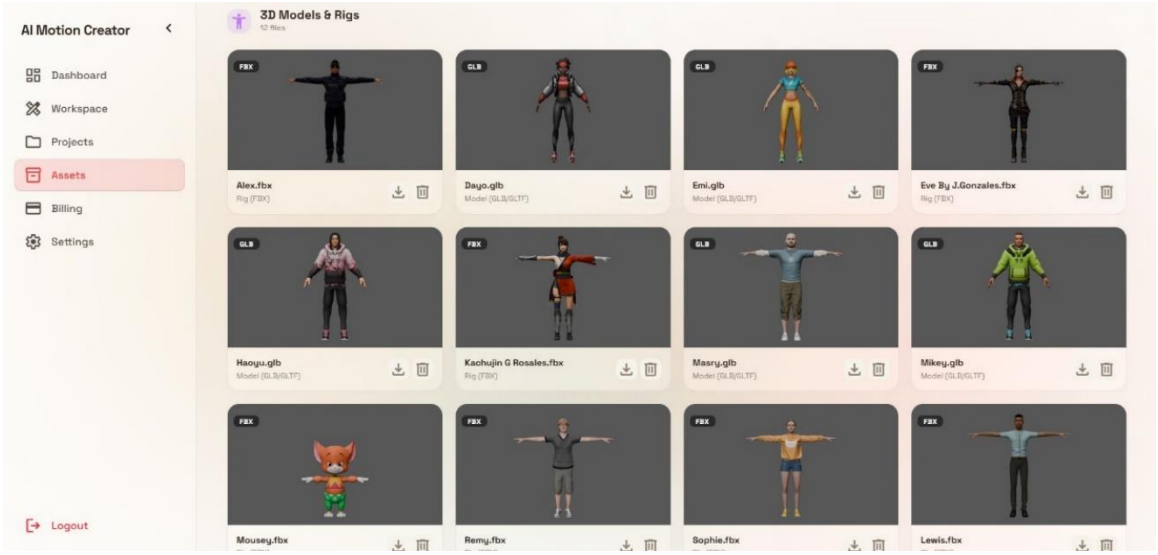


Figure 5.12, Assets library: 3D models and rigs available for retargeting.

Settings. The Settings page (Figure 5.13) manages the account and preferences: profile, account security (password change and two-factor authentication), appearance (light/dark/blue theme and accent colour), and notification toggles.

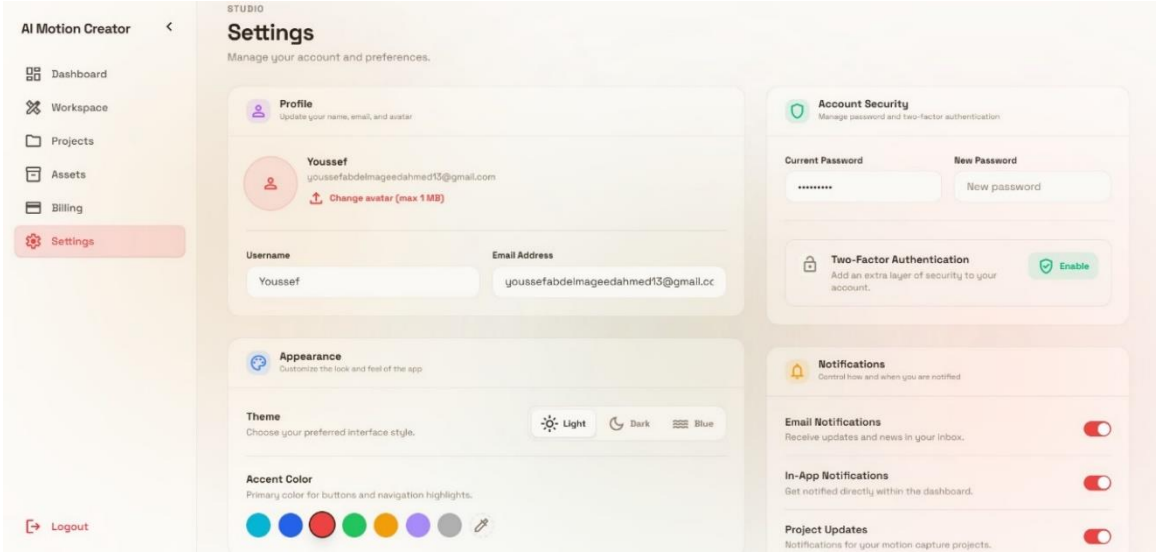


Figure 5.13, Settings: profile, security, appearance, and notifications.

5.3 Results and Discussion

The finished platform was tested at three levels, performance under load, processing speed of the motion pipeline, and functional completeness, to establish what it achieves in practice. This section reports the measured results and discusses what they mean.

5.3.1 Results

Functional completeness. All of the platform's core features work end to end. Registration and login (including multi-factor sign-in), video upload, motion extraction, export to BVH, FBX, and GLB, retargeting onto a custom rig, 3D preview, AI text/image-to-video generation, file and project management, search and filtering, the simulated billing flow, and the per-user dashboard all completed successfully during unit, integration, and user-acceptance testing.

Performance under load. The deployed web and data layer were measured under increasing concurrent load. Everyday read interactions stayed fast and error-free, holding average response times below roughly half a second up to about one hundred concurrent users, with no failed requests, and degrading gracefully rather than failing as load was pushed further. This confirms that the managed front-end and database layer comfortably handle the concurrency expected at project scale.

Processing speed of the pipeline. The motion pipeline's two stages behaved very differently, as summarised in Table 5.4. Skeleton (BVH) extraction was fast, a few seconds for a short clip. The rigged-model conversion stage (FBX/GLB), which runs the headless 3D engine, was far heavier, taking on the order of a few minutes for a short clip and, for longer clips on the cloud tier, able to exceed the hosting gateway's request-time limit. This stage is the platform's main performance bottleneck and runs most reliably on the local or desktop configuration.

Table 5.4, Measured processing behaviour of the motion pipeline.

Pipeline Stage	Short Clip	Notes
Skeleton extraction (BVH)	A few seconds	Fast. Suitable for the cloud tier.
Rigged-model export (FBX / GLB)	Order of a few minutes	Heavy 3D-engine stage. Can exceed the cloud request limit on longer clips. Best run locally.

5.3.2 Discussion

The results show that the central goal of the project was achieved: an ordinary 2D video can be turned into a usable, retargeted 3D character animation through an accessible web platform, without specialist hardware. The web and data layer proved fast and robust under load, demonstrating that an affordable managed-cloud deployment is sufficient for the platform's interactive features.

The clearest limitation is the cost of the rigged-model conversion stage. Because it drives a full 3D engine, it is computationally expensive and can exceed the request-time limit of the cloud host on longer clips. This is why the platform offers the lighter skeleton export on the cloud and reserves the heavy conversion for the local and desktop configurations, and it is the primary target for future optimisation. A second limitation is that motion accuracy is bounded by what single-camera video allows: depth ambiguities and occlusions cannot be fully resolved from one viewpoint, so the system targets plausible, usable motion rather than studio-grade precision. Finally, the billing flow is simulated rather than connected to a live payment provider, which is the main outstanding gap before any commercial launch. Despite these limitations, the platform delivers a complete, secure, and easy-to-use path from video to 3D animation, and the measured results confirm that its design choices are sound.

Chapter 6: Testing

Testing was carried out to confirm that the AIMotionCreator platform behaves correctly, both at the level of its individual components and when those components are combined into complete workflows. This chapter presents the testing in two parts: unit testing, which exercises the smallest pieces of the system in isolation, and integration testing, which verifies that the separate parts work correctly together. Each part is supported by a table of test cases giving the expected and actual outcomes.

6.1 Unit Testing

Unit testing focused on the smallest pieces of the system in isolation. On the AI side, this meant confirming that body and hand tracking returned the expected joint positions on sample frames, that the angle and rotation calculations produced correctly oriented skeletons, and that the resulting motion file was written in a valid format. On the front-end side, individual interface components and form checks were exercised on their own, confirming, for example, that invalid email addresses and empty required fields were rejected before a form could be submitted. Table 6.1 lists the principal unit test cases and their results.

Table 6.1, Unit test cases and their results.

ID	Test Case	Expected Result	Result
UT-01	Run pose detection on a sample video frame.	Body joint positions are returned for the visible subject.	Pass
UT-02	Run hand-landmark detection on a sample frame.	Finger and hand joint positions are returned.	Pass
UT-03	Compute joint angles and rotations from tracked points.	A correctly oriented skeleton is produced.	Pass
UT-04	Write the extracted motion to a BVH file.	A valid, well-formed BVH file is produced.	Pass

UT-05	Submit the sign-up form with an invalid email address.	The form is rejected with a validation message.	Pass
UT-06	Submit a form with a required field left empty.	Submission is blocked and the field is flagged.	Pass
UT-07	Change the theme and reload the application.	The chosen theme persists across the reload.	Pass
UT-08	Calculate the credit cost for a clip of known duration.	The reported cost matches the processed length.	Pass
UT-09	Apply the asset filter to a library with hidden/deleted items.	Hidden and deleted items are excluded from the list.	Pass
UT-10	Sign in to an account with multi-factor authentication enabled.	A second verification step is presented before access.	Pass

All unit test cases passed. Each component behaved as specified in isolation, giving confidence that the building blocks of the system were correct before they were combined.

6.2 Integration Testing

Integration testing verified that the separate parts of the system worked correctly together. The complete motion pipeline was exercised end-to-end, uploading a video, extracting the motion, converting it to each export format, retargeting it onto a character, and previewing the result in the browser. The exchange between the front-end and the processing server was tested for the correct handling of uploads, settings, and downloads, and the reading and writing of user data and files to the cloud database and storage was confirmed. Finally, the hybrid cloud deployment was tested as a whole, ensuring that the hosted front-end, the hosted back-end, and the secure tunnel to the local language model all operated together. Table 6.2 lists the principal integration test cases and their results.

Table 6.2, Integration test cases and their results.

ID	Test Case	Expected Result	Result
IT-01	Upload a video and generate skeleton motion (BVH).	Motion is extracted and a valid BVH is produced.	Pass
IT-02	Export a rigged model (FBX / GLB) via the 3D engine.	A valid rigged model is produced.	Pass *
IT-03	Retarget the captured motion onto a custom rig.	The motion plays correctly on the custom character.	Pass
IT-04	Load the generated model in the in-browser 3D viewport.	The animation previews interactively in the browser.	Pass
IT-05	Exchange uploads, settings, and downloads between front-end and back-end.	Requests and responses are handled correctly across domains.	Pass
IT-06	Register, log in, complete MFA, and reset a password.	Authentication flows complete through the cloud auth service.	Pass
IT-07	Create, read, update, and delete assets and projects.	Changes persist to the database and storage and reload correctly.	Pass
IT-08	Search and filter the assets and projects library.	The correct subset of items is returned.	Pass
IT-09	Generate a video with the text-to-video tool.	A clip is generated through the external service and shown inline.	Pass
IT-10	Run the full hybrid deployment (cloud front-end, cloud back-end, local model tunnel).	All three layers operate together over the internet.	Pass
IT-11	Submit the public contact form.	The message is delivered by email through the email service.	Pass

All integration test cases passed. One case is marked with an asterisk (*): the rigged-model export (IT-02) produces correct output, but the 3D-engine conversion stage is computationally heavy and, for longer clips on the cloud tier,

can exceed the hosting gateway's request-time limit. It therefore completes most reliably on the local or desktop configuration. This is a performance constraint rather than a correctness failure, and it is identified as the primary target for future optimization. With this caveat noted, the integrated platform was confirmed to work correctly from end to end across all of its main workflows.

6.2.1 Sample Input and Output

To illustrate the end-to-end integration test, the figures below show a representative input and the output it produced. Figure 6.1 shows the input stage, where a 2D source video is selected and uploaded for processing.

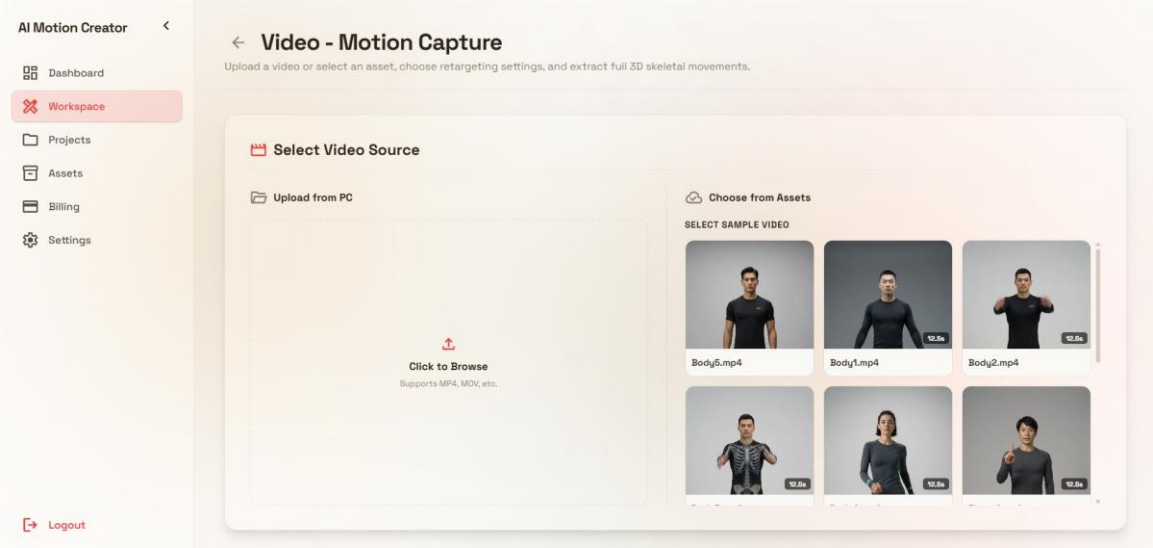


Figure 6.1, Integration test input: selecting and uploading the 2D source video.

Figure 6.2 shows the generation controls used to configure the test run, the export type, the motion-solver options, and the optional custom rig, before the pipeline is started.

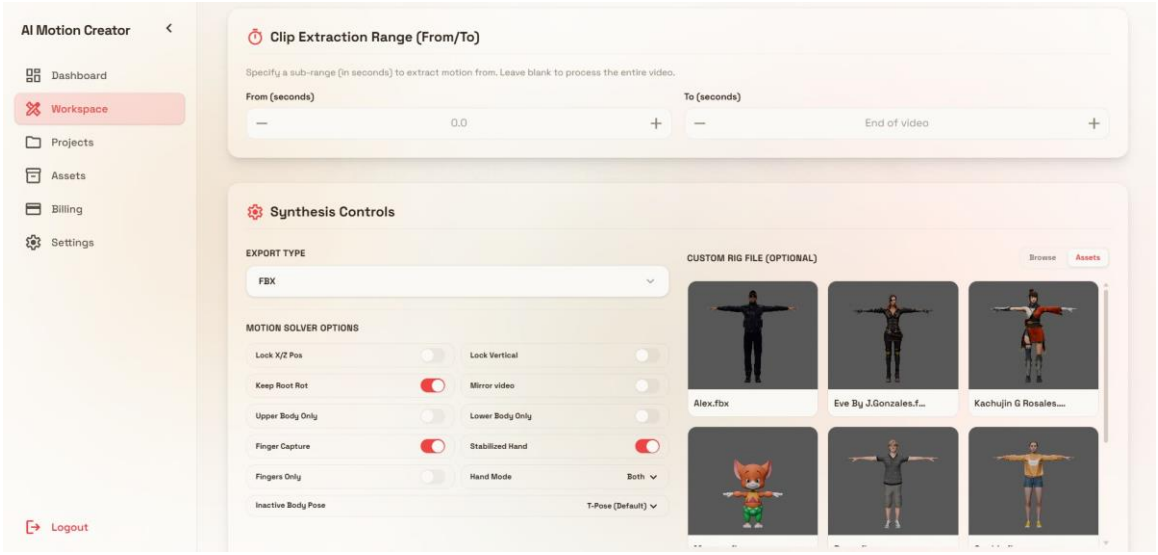


Figure 6.2, Integration test configuration: export type and motion-solver options.

Figure 6.3 shows the output stage: the generated 3D animation displayed in the interactive viewport beside the source video, confirming that the input was processed correctly through the full pipeline.

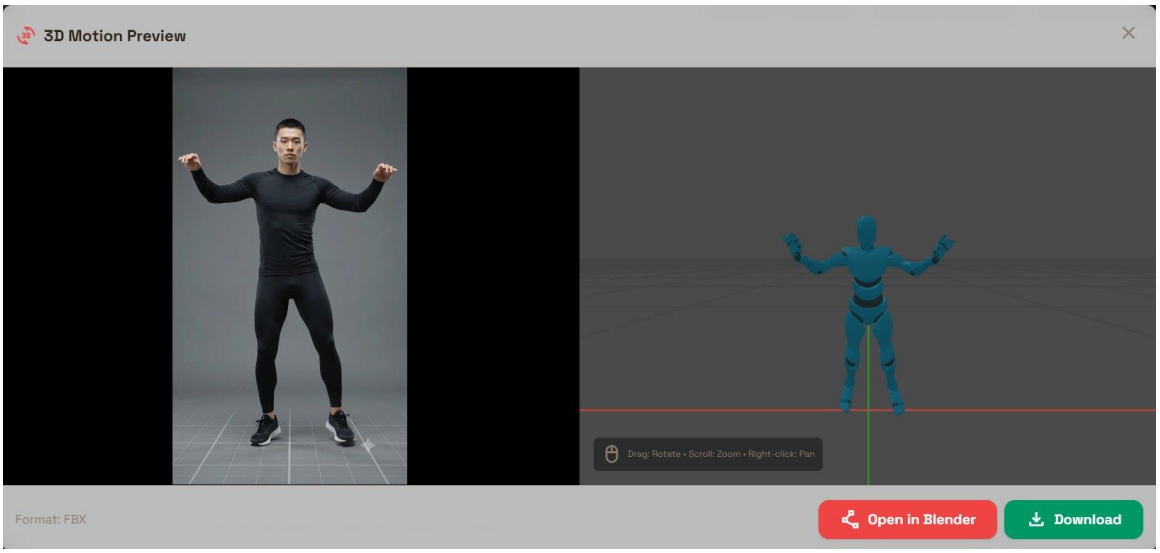


Figure 6.3, Integration test output: the generated 3D animation beside the source video.

Figure 6.4 shows the generated file recorded in the Projects library, confirming that the output was correctly saved and made available for download.

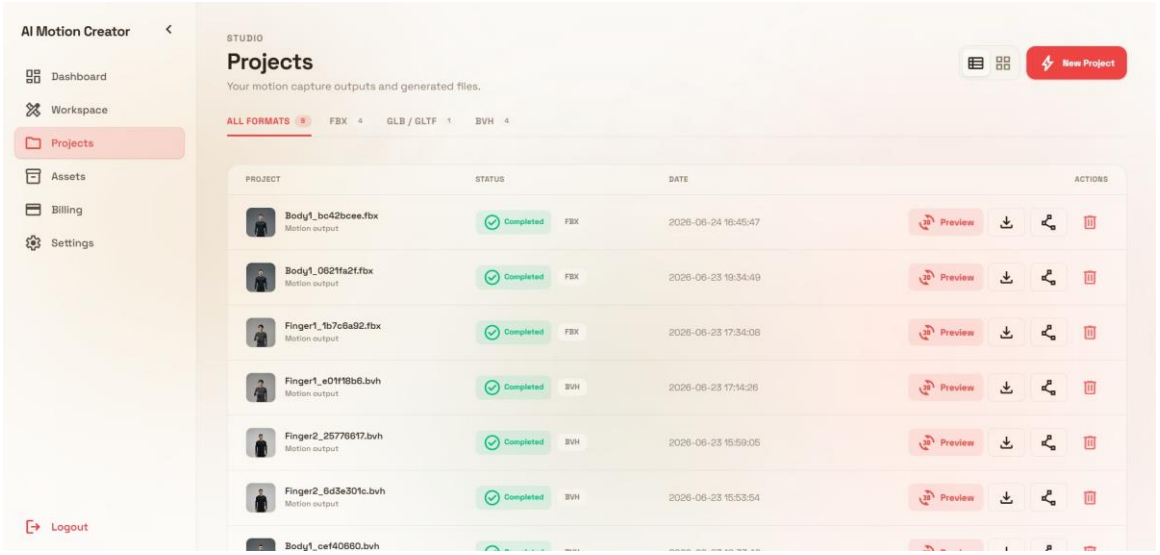


Figure 6.4, Integration test output: the generated file saved in the Projects library.

Chapter 7: Conclusion and Future Work

Conclusion

The aim of this project was to make 3D character animation accessible without specialist motion-capture hardware or expert 3D skills, by turning an ordinary 2D video into a usable, retargeted 3D character animation through a simple web interface. The platform delivers on that aim: a user can upload a normal video, have the motion extracted automatically, choose an export format, optionally retarget the motion onto a custom character, preview the result in an interactive 3D viewport, and download or open it for further work, all from the browser, and also from an offline desktop version built from the same core.

The objectives set out at the start of the project were each met. An AI pipeline was built to reconstruct 3D motion from a single-camera video using modern body- and hand-tracking with a 3D conversion and retargeting stage. This pipeline was wrapped in an approachable web application backed by a cloud database, authentication, and storage service. AI-assisted text-to-video and image-to-video tools were added, and the system was deployed on a hybrid cloud-and-local architecture that keeps the heavy language model on local hardware while serving the rest of the platform from the cloud.

Testing and evaluation confirmed that these objectives translate into a working, usable product. Unit and integration testing showed that every specified component and workflow passed. Performance testing against the live deployment showed that the web and data layer is fast and reliable under heavy concurrent load and that common skeleton extraction completes within seconds, while identifying the rigged-model conversion stage as the main performance bottleneck. The project makes several practical contributions: an end-to-end, no-hardware motion-capture pipeline. A unified, accessible web platform built around it with accounts, storage, preview, and export. A working hybrid cloud-

and-local architecture that deploys an AI application affordably, and the integration of generative AI tools beside a traditional production pipeline. In short, the project succeeded in taking a complex, hardware-dependent task and making it available to anyone with a video and a web browser.

Future Work

The limitations identified during development and evaluation translate directly into a clear, actionable agenda for future development. The most impactful items address the bottleneck identified in testing, followed by features that would broaden the platform's reach and value:

- Move heavy jobs onto a background queue with multiple workers. Running the 3D conversion stage as a queued background task spread across several processing workers, rather than inside a single web request, would remove the gateway time-limit problem and let the cloud service handle many simultaneous rigged-model exports.
- Host the language model on always-available infrastructure. Replacing the local-machine-and-tunnel arrangement with a hosted model would make the text-to-video feature reliable around the clock and independent of any single computer being switched on.
- Integrate a real payment gateway. Connecting the existing simulated billing flow to a real payment provider would turn the subscription tiers into a working commercial offering with automated credit top-ups and plan upgrades.
- Improve motion accuracy with multi-view or depth input. Supporting more than one camera angle, or optional depth input, would improve reconstruction of occluded and fast movements and reduce the estimation error inherent in single-camera capture.

- Add a dedicated mobile application. A mobile app would let users capture and upload motion video directly from a phone and review results on the go, widening access beyond the desktop browser.
- Add an AI animation assistant (chatbot). An in-app conversational assistant could guide first-time users, recommend the best capture and export settings for a given clip, and answer questions.

Implemented together, these directions would evolve the platform from a successful proof of concept and working product into a robust, fully scalable, and commercially ready service, while preserving the core achievement of this project: making 3D character animation accessible to anyone, from a single ordinary video.

References

[1] Science and Media Museum, “The surprising origins of motion capture.” https://www.scienceandmediamuseum.org.uk/objects-and-stories/surprising-origins-motion-capture

[2] Fleischer Studios, “The Rotoscope.” https://www.fleischerstudios.com/mrotoscope.html

[3] Engadget, “Motion capture's long road to Hollywood.” https://www.engadget.com/2018-05-25-motion-capture-history-video-vicon-siren.html

[4] T. Mocap, “Motion capture (thesis),” Theseus. https://www.theseus.fi/bitstream/10024/336908/2/taryn_mocap_thesis.pdf

[5] Encyclopaedia Britannica, “Motion capture.” https://www.britannica.com/technology/motion-capture

[6] Los Angeles Times, “How the ‘Avatar’ sequel takes its technology underwater.” https://www.latimes.com/entertainment-arts/awards/story/2023-01-02/how-avatar-sequel-takes-its-technology-underwater-for-an-immersive-experience

[7] Vicon, “How motion capture continues to revolutionize game development.” https://www.vicon.com/resources/blog/how-motion-capture-continues-to-revolutionize-game-development/

[8] Wikipedia, “Motion capture.” https://en.wikipedia.org/wiki/Motion_capture

[9] Motion Capture Society, “Industry history.” http://www.motioncapturesociety.com/resources/industry-history

[10] Roboflow, “The history of pose estimation algorithms.” https://blog.roboflow.com/pose-estimation-algorithms-history/

[11] DhiWise, “MediaPipe vs OpenPose: a practical guide to pose analysis.” https://www.dhiwise.com/post/mediapipe-vs-openpose-a-practical-guide-to-pose-analysis

[12] arXiv:1604.03265. https://arxiv.org/abs/1604.03265

[13] arXiv:2402.19144. https://arxiv.org/abs/2402.19144

[14] Theia Markerless, “Camera system requirements.” https://docs.theiamarkerless.com/theia3d-documentation/camera-system-requirements

[15] Theia Markerless, “Budgeting for a markerless system.” https://www.theiamarkerless.com/blog/budgeting-for-a-markerless-system

[16] Theia Markerless, demonstration video. https://youtu.be/s0A-hvWvH5g

[17] Vicon, “Nexus, Theia.” https://www.vicon.com/software/nexus/theia/

[18] Vicon, “What is markerless motion capture?” https://www.vicon.com/resources/blog/what-is-markerless-motion-capture/

[19] The Captury, “Post-processing.” https://captury.com/post-processing/

[20] The Captury, “Real-time processing.” https://captury.com/real-time-processing/

[21] Simi Reality Motion Systems, “Markerless motion capture.” http://www.simi.com/en/products/movement-analysis/markerless-motion-capture.html

[22] Radical Motion, “Best motion capture software 2025.” https://radicalmotion.com/changelog/insights/best-motion-capture-software-2025

[23] Move.ai, “Knowledge base.” https://docs.move.ai/knowledge

[24] Rokoko, “Introducing Rokoko Vision.” https://www.rokoko.com/insights/introducing-rokoko-vision-elevate-your-motion-capture-with-dual-camera-precision

[25] Autodesk, “Flow Studio overview.” https://www.autodesk.com/products/flow-studio/overview

[26] DeepMotion, “Animate 3D documentation.” https://www.deepmotion.com/doc/animate-3d

[27] Skywork, “Quick Magic in-depth review (2025).” https://skywork.ai/skypage/en/Quick-Magic-In-Depth-Review-2025/1976182381206368256

[28] Plask, “Documentation.” https://plask.ai/en-US/docs

[29] NVIDIA, “Maxine AR SDK system guide.” https://docs.nvidia.com/deeplearning/maxine/ar-sdk-system-guide/index.html

[30] Google, “MediaPipe Solutions guide.” https://ai.google.dev/edge/mediapipe/solutions/guide

[31] FreeMoCap, GitHub repository. https://github.com/freemocap/freemocap

[32] CMU Perceptual Computing Lab, “OpenPose,” GitHub. https://github.com/CMU-Perceptual-Computing-Lab/openpose

[33] Viso.ai, “OpenPose.” https://viso.ai/deep-learning/openpose/

[34] Google MediaPipe, “MediaPipe Solutions: Pose Landmarker.” https://developers.google.com/mediapipe

[35] Blender Foundation, “Blender Documentation.” https://docs.blender.org

[36] OpenCV, “Open Source Computer Vision Library.” https://opencv.org

[37] FastAPI Documentation. https://fastapi.tiangolo.com

[38] Next.js Documentation. https://nextjs.org/docs

[39] Three.js Documentation. https://threejs.org/docs

[40] NumPy Documentation. https://numpy.org/doc

[41] SciPy Documentation. https://docs.scipy.org

[42] Blender, “BVH animation format.” https://docs.blender.org/manual/en/latest/addons/import_export/anim_bvh.html

[43] Adobe, “Mixamo character animation platform.” https://www.mixamo.com

Appendices

Appendix A: GitHub Repo

GitHub Link: https://github.com/xYoussefHelalx/AIMotionCreator

ملخص المشروع باللغة العربية

يُعد AIMotionCreator منصةً قائمةً على الويب تجعل تحريك الشخصيات ثلاثية الأبعاد في متناول المستخدمين دون الحاجة إلى أجهزة التقاط حركة متخصصة أو خبرة فنية. تعتمد المنصة على الذكاء الاصطناعي ورؤية الحاسوب لإعادة بناء حركة الجسم واليدين من مقطع فيديو ثنائي الأبعاد عادي وتحويلها إلى صيغ تحريك ثلاثية الأبعاد قياسية، يمكن إسقاطها على نماذج شخصيات مخصصة ومعاينتها في عارض تفاعلي ثلاثي الأبعاد وتصديرها للاستخدام في أدوات أخرى.

بُنيت المنصة على معمارية ثلاثية الطبقات: واجهة أمامية للويب باستخدام Next.js، وخادم خلفي باستخدام FastAPI ينسّق مسار ذكاء اصطناعي يعتمد على MediaPipe وBlender، وقاعدة بيانات مُدارة عبر Supabase توفّر البيانات والمصادقة والتخزين.

كما تدمج المنصة أدوات توليدية لتحويل النص إلى فيديو والصورة إلى فيديو، وتُنشر عبر معمارية هجينة منخفضة التكلفة تجمع بين السحابة والتشغيل المحلي، مع توفّر نسخة مكتبية تعمل دون اتصال. خضعت المنصة المكتملة للتقييم من حيث الأداء وقابلية الاستخدام واكتمال الوظائف. أظهر اختبار الأداء على النسخة المنشورة أن طبقة الويب والبيانات ظلت سريعة وخالية من الأخطاء تحت حِمل متزامن كبير، مع تحديد مرحلة تحويل النموذج المُجهَّز بالهيكل العظمي بوصفها عنق الزجاجة الرئيسي في المعالجة.

وقد تحققت جميع المتطلبات الوظيفية. يُثبت المشروع أن التقاط الحركة دون علامات وبطريقة ميسّرة يمكن تقديمه كتطبيق ويب متكامل وآمن وسهل الاستخدام.

يهدف المشروع إلى إزالة حواجز التكلفة والأجهزة والخبرة التي حصرت تحريك الشخصيات ثلاثية الأبعاد تقليديًا في الاستوديوهات الكبيرة، وإتاحته للمطورين المستقلين وطلاب التحريك وصناع المحتوى والاستوديوهات الصغيرة. ومن أبرز ما توصل إليه المشروع أن طبقة الويب والبيانات تعمل بكفاءة وموثوقية عالية، بينما تظل مرحلة تحويل النموذج المُجهَّز بالهيكل العظمي هي الأكثر استهلاكًا للموارد، وهي المرشّح الأول للتحسين مستقبلًا إلى جانب ربط نظام الدفع الفعلي وتحسين الدقة باستخدام عدة زوايا تصوير.